Progress Report

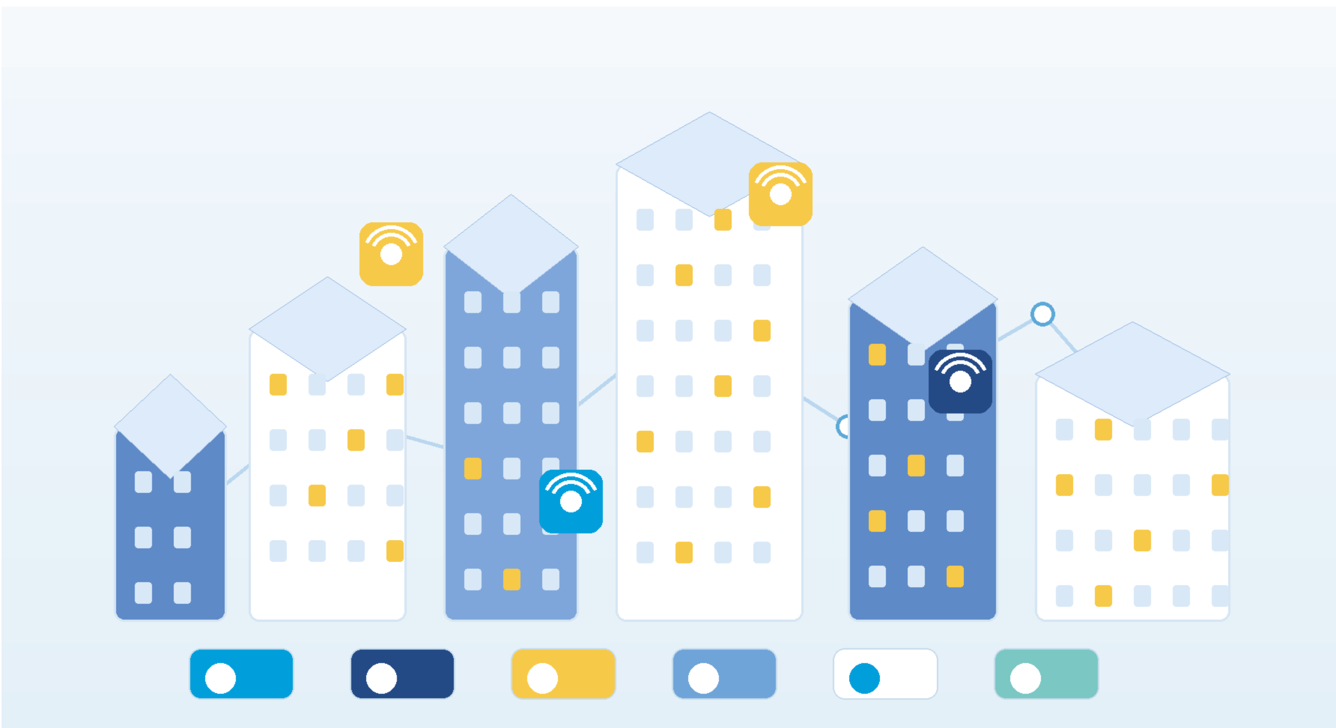
ITU UMC Data Hackathon 2026

Seoul Integrated Connectivity Analysis

Measuring Universal and Meaningful Connectivity

in a Hyper-Connected City

An Integrated Methodology for Measuring UMC Across Seoul's Districts



Integrated Methodology for Seoul's 25 Districts

May 21 2026

Key Results

The project integrates district indicators, multilevel survey analysis, and platform–visible signals to measure UMC in Seoul without reducing local connectivity to a single infrastructure metric.

01 District UMC gaps

Seoul shows meaningful district–level disparities across the six UMC dimensions; the composite index ranges from Seocho–gu at 0.695 to Jungnang–gu at 0.279.

02 Individual vulnerability dominates

The HLM results indicate that digital–use inequality is structured primarily by individual characteristics; the unconditional ICC is 0.49%.

03 Place still matters for targeting

District context is not treated as a broad causal average, but as a place–sensitive targeting layer that helps locate where support should be adjusted.

04 Platform text adds lived–experience signals

LLM–assisted classification is used as structured coding for exploratory signals, not as direct prevalence estimation or unconstrained model interpretation.

05 Bayesian updating links layers

Measured district UMC scores are used as prior information, and classified platform signals are used to inspect posterior shifts by district and dimension.

06 Spatial matching was tightened

All 131,792 classified rows were matched to administrative–neighborhood units before district aggregation, with living–population linkage used for consistent place–time weighting.

Interpretive boundary These front–matter results summarize the report–level evidence. They do not expose raw platform text, post IDs, or private examples, and they treat text–derived signals as exploratory evidence requiring cautious interpretation.

1. Overview

1.1 Theoretical Understanding

This project examines Seoul's digital connectivity through three analytical stages and derives policy recommendations on the basis of the results. Digital connectivity is grounded in the Universal Meaningful Connectivity (UMC) framework proposed by the ITU. The project's analyses are interpreted not as a juxtaposition of separate outputs, but within a single metatheoretical framework. Metatheory, as a comprehensive view of the nature of reality, provides philosophical support for research and practice (Allana & Clark, 2018), and offers a basis for specifying the ontological position of each methodology. By explicitly showing the process through which observed phenomena are generalized into concepts at different Ontological levels, the project seeks to establish its own validity. This approach has the advantage of providing not only validity for the project itself, but also a clear basis for application in other contexts.

It can offer alternative points of reference for **current policy implementatio**n. As evidence-based policy has become widespread, there have been increasing cases in which analytical results are borrowed and applied without sufficient consideration of the institutional and cultural conditions of the original context (Dolowitz & Marsh, 2000). At the same time, critiques of the evidence-based paradigm itself have accumulated. A representative critique is that the narrow evidence-based framework fails to capture the complex, context-dependent, and value-laden character of policymaking (Greenhalgh & Russell, 2009). Accordingly, a transition from evidence-based policy to evidence-informed policy has been discussed (Head, 2010), and the role of evidence is being redefined as guiding policy rather than constituting its systematic foundation. Rather than diminishing the importance of empirical evidence, this project aims to clarify in detail the conditions and core structures required for applying evidence to other contexts on the basis of a theoretical framework. It explicitly presents the role of each analysis, their interconnections, and the interpretive structure that leads to policy recommendations, while also specifying the key contextual information needed for interpretation. This reflects the case-study character of the project itself, which is distinct from the individual analytical methods.

The individual analytical methodologies are divided into three types. First, the theoretical framework of UMC is applied to Seoul's 25 districts, which are the actors responsible for policy implementation, in order to measure relative disparities. Next, multilevel models are used to separate the contributions of individual factors and district-level contextual factors to individuals' digital connectivity problems. Finally, platform-visible evidence on Seoul's digital connectivity problems, as generalized in the preceding analyses, is reconnected to individual phenomena observed through unstructured text. This includes a procedure for abstracting and reconstructing individual experiences in accordance with the conceptual framework of UMC. The concept of UMC proposed by the ITU is compared with aggregated results and platform-visible lived-experience signals. Finally, plausible pathways through which actual phenomena may occur are interpreted from the concept. These transitions across levels are grounded in the stratified ontology proposed by critical realism, namely the distinction among the empirical, actual, and real domains (Bhaskar, 1975).

The need for this process can be illustrated through a hypothetical situation. Suppose, for example, that observed empirical evidence shows that residential areas in station areas served by express subway lines have higher housing prices than residential areas around stations that are not served by such lines. Apart from the validity of the result, the finding itself is difficult to regard as meaningful knowledge. It may apply only to Seoul. Moreover, making every station served by an express line on the basis of this result is neither politically nor logically feasible. In the preceding situation, one example would be to identify a mechanism whereby the high value of time in Seoul causes express lines that reduce travel time to be reflected in housing prices. The next section explains the advantages that the individual analytical methodologies themselves offer for application in other contexts.

1.2 Understanding the Analytical Methods

The purpose of this section is to explain the three analytical methods and, separately from the metatheoretical framework presented above, to clarify the basis on which each methodology itself can be applied to other contexts. First, empirical patterns are identified through regression-based analysis of the data. The discussion separates the validity of the metalevel structure, which measures and abstracts individual observations and then restores them to individual phenomena while testing whether the UMC concept is also manifested empirically, from the applicability of the individual methodologies.

First, district-level UMC in Seoul is measured to show where disparities exist and along which dimensions. The measurement indicators are based on the conceptual definitions of the six dimensions proposed by ITU (2023), but are operationalized to fit Seoul's data environment and urban characteristics. The actors that actually implement policy are distributed not only at the national level but also across various units, including local governments. Accordingly, this project sets Seoul's 25 districts as the unit of measurement and analysis so that the analytical unit corresponds to the unit of policy implementation. Because the conceptual definitions and the operational definitions appropriate to the study area are applied separately, indicators can be substituted to fit each city's data environment while maintaining the same six-dimensional structure. The selection of measurement indicators and the results are detailed in Section 3.1.

Next, using a two-level hierarchical linear model (HLM), a model that separates individual-level and district-level factors in nested data, the analysis examines for whom and under what contextual conditions differences in digital usage are most pronounced. The aim is to conduct multilevel association analysis rather than causal inference. Even within the same administrative area, digital usage can vary substantially according to individuals' educational attainment, age, and household composition. HLM is a standard method for accounting for the hierarchical data structure in which individuals are nested within districts, and the same analytical framework can be applied to other cities by replacing the input variables according to the demographic characteristics and infrastructure conditions of the relevant context. The selection of variables, model specification, and detailed design of cross-level interactions are presented in Section 3.2.

Finally, unstructured text is used to identify platform-visible lived-experience signals that the preceding two analyses cannot capture and to interpret them in relation to the UMC framework. Individual cases are not immediately equated with the district-level UMC concept; rather, they pass through a structured coding and interpretation process and are compared with the preceding observed results in order to illustrate plausible experiential pathways. This procedure does not prove the multilevel model or estimate population prevalence. Its value is to make visible how administrative and survey-based patterns may appear in everyday situations.

Text analysis has two requirements. First, it must be possible to determine whether individual text cases are related to digital connectivity and to classify them according to the predefined theoretical framework of the six UMC dimensions. Second, the analytical method must not be strongly tied to a particular language. Because the UMC framework is comprehensive, text analysis likewise requires language independence. To satisfy these two conditions, a large language model (LLM), used here as a structured coding assistant for classifying platform text against predefined criteria, is applied as a theory-driven text classification tool. Because an LLM processes meaning at the contextual level rather than at the morpheme level, it is robust to nonstandard expressions in Korean everyday-life text; by directly presenting the UMC dimensions in the prompt, the researcher can control the classification criteria. Methodological transferability is secured insofar as the same prompt structure can be applied to texts in other languages, although this does not replace human validation.

Specifically, the measured UMC values from Section 3.1 are used as prior information and the text classification results are treated as platform-visible signals. Bayesian updating, a method for revising prior information with new evidence, is used here as an exploratory evidence-integration procedure rather than as a confirmatory estimate of deprivation. Then, within the cautious interpretive range permitted by the HLM results in Section 3.2, the relationship between structural indicators and lived experience is visualized. Finally, plausible pathways between individual posts and the district-level UMC concept are interpreted. At this stage, three independent hypothesis agents are designed-abductive, forward-reasoning, and sequential-so that each agent generates hypotheses without referring to the outputs of the others, after which a judgment synthesizer integrates them under an expanded information set. Separating the inference layers suppresses bias from any single inferential path and makes it possible to observe convergence and divergence among hypotheses. The prompt design, strategies for reducing misclassification, Bayesian updating, and detailed inference procedures are presented in Section 3.3.

In summary, within the structure of this project, Chapter 2 provides the interpretive context for reading the results in Chapter 3, and Section 3.3 functions as a mediating stage that reconnects the results of Sections 3.1 and 3.2 with experiences in the lifeworld. The three analyses answer different questions, while forming a sequential structure in which the output of one analysis becomes the input for the next analysis (see Figure 1). By separating conceptual structure from operational definitions and specifying which elements are replaceable and which are fixed, each methodology establishes the conditions under which the same analytical framework can be applied beyond Seoul (Figure 1).

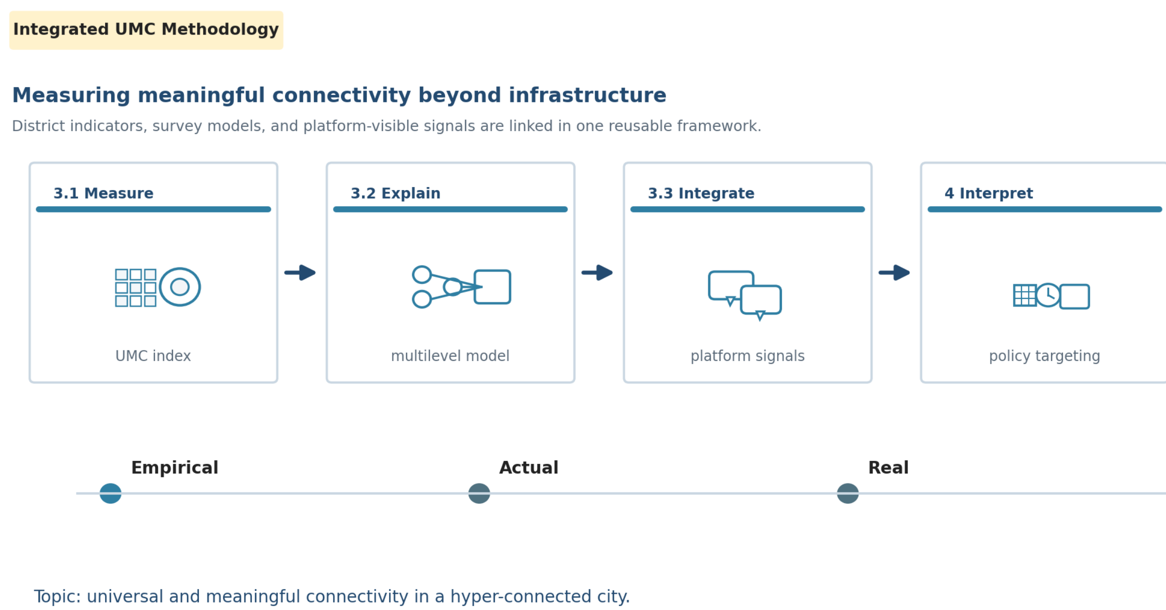


Figure 1. Integrated UMC Measurement Framework for a Hyper-Connected City

2. Theoretical Context

This chapter presents the theoretical foundations for the analysis and the interpretive context for Chapter 3. It first reviews the individual and regional factors that affect the digital connectivity. It then examines the distinction in public policy between a people-based approach and a place-based approach, as well as the background of recent discussions on a place-sensitive approach. Next, it reviews critical realism, the metatheoretical framework of this project, and cases of its application in empirical research. Finally, it examines Seoul's demographic, administrative, and infrastructure conditions in order to provide the contextual information needed to interpret the analytical results.

2.1 Universal Meaningful Connectivity

Universal Meaningful Connectivity (UMC) is a policy goal that all people should be able to use the internet in a safe, satisfying, enriching, and productive way at an affordable cost (International Telecommunication Union, 2025). This study develops that goal from a critical realist perspective by reviewing two conceptual strands. The first concerns the capability-based character of UMC indicators, and the second concerns the gap between collective UMC and individual digital connectivity.

The first strand begins with Sen's capability approach. Sen (1985) argued that well-being should be evaluated not by the resources people possess in themselves, but by what people are actually able to do and to be. Subsequent capability research has emphasized that the same resources can be converted into different real opportunities depending on personal conditions, social arrangements, and institutional environments (Robeyns, 2005). This logic is directly connected to the ITU's concept of universal and meaningful connectivity. Mere network access or device ownership is not sufficient; people must also be able to convert those resources into meaningful use in everyday life. A capability-based approach also raises the question of thresholds, namely what counts as a minimum socially acceptable level. In low-connectivity settings, this threshold is often expressed primarily as the presence or absence of basic internet access.

Seoul, however, requires a different operationalization of the threshold idea because it already has advanced digital infrastructure and a high level of smartphone diffusion. In the original UMC framework, threshold-based measurement is appropriate for identifying whether a socially acceptable minimum level of meaningful connectivity has been reached. In Seoul, many basic access thresholds have already been passed, so applying only a threshold criterion would make it difficult to observe meaningful differences among districts. The empirical strategy therefore retains the normative concern of the threshold approach, but operationalizes it through within-Seoul Min-Max normalization in order to compare the relative positions of the 25 districts on a common scale. The threshold problem thus moves from the presence or absence of basic access toward the relative quality, affordability, usability, and security of digital connection in ordinary urban life.

The second strand concerns the conceptual gap between UMC at the national or regional level and digital connectivity at the individual level. UMC was originally formalized as a national-level policy and measurement concept, and the ICT Development Index evaluates how universal and meaningful connectivity is at an aggregate scale (International Telecommunication Union, 2025). Individual digital connectivity, however, cannot be reduced to a simple arithmetic sum of national or district-level indicators. Research on digital inequality shows that even after physical access expands, differences persist in skill, forms of use, outcomes, and social position (Hargittai, 2002; Helsper, 2012; van Deursen & van Dijk, 2015; Scheerder et al., 2017). In other words, people can have different digital experiences even under the same regional conditions.

This gap provides the conceptual basis for the multilayered design of this study. The reason this study adopts critical realism is that district-level indicators, individual survey responses, and platform posts do not measure the same phenomenon at merely different scales. Rather, they correspond to different layers of reality: structural conditions, individual use, and lived events. The Bayesian updating used later in the analysis provides a controlled but exploratory way to connect these layers without reducing one layer to another. District-level UMC scores provide prior information on contextual conditions, while classified platform posts provide platform-visible signals about how deprivation is narrated and made visible. Therefore, posterior shifts are not interpreted as increases in complaint prevalence or as confirmation of local deprivation; they are interpreted as signals that help examine the relationship between collective UMC conditions and observed lived experience.

2.2 Policy

In the design of public policy, the distinction between interventions that target people's characteristics, or a people-based approach, and those that target place conditions, or a place-based approach, is a long-standing debate. People-based policies seek to improve outcomes by directly supporting human characteristics such as individual competency, income, and education. Place-based policies seek to indirectly improve outcomes for residents of a given area by improving that area's infrastructure, institutions, and environmental conditions (Neumark & Simpson, 2015). The two approaches are not mutually exclusive, but because the allocation of policy resources and the pathways of intervention differ, they need to be designed separately from the perspective of policy implementation.

2.2.1 Factors Affecting the Digital Divide: Individual Characteristics and Regional Conditions

In the literature on the digital divide, studies that focus on individual characteristics show that, even within the same region, individual characteristics are a principal factor explaining disparities in digital usage. Hargittai (2010) argued that the divide after internet access-that is, differences in patterns of use-determines substantive inequality. This finding suggests that disparities in digital usage may persist according to socioeconomic status. Chipeva et al. (2018) showed that usefulness is the strongest motivation for technology use across various countries, while also demonstrating that individuals' internal characteristics significantly affect the use of digital devices. Borges et al. (2020), through the case of Angola, reported that habits and proficiency in ICT skills are key factors in the divide prior to the diffusion of devices. Hilbert's (2011) analysis of 18 Latin American countries showed that a gender digital divide exists and that internet use is more active in countries with higher levels of gender equality.

At the same time, evidence has also accumulated that local environments have additional explanatory power for digital usage even after controlling for individual characteristics. Applying HLM to the United States, Mossberger et al. (2006) showed that disparities in internet access are associated with high-poverty neighborhood environments rather than race itself. This suggests that local structural conditions may shape the context in which individual outcomes are produced. Through a comparative qualitative study of local digital-inclusion strategies in the United Kingdom, Branstrator-Erickson & Ellison (2023) also argued that regional digital-inclusion policies cannot be explained solely by individual-level factors; they are additionally shaped by the interorganizational governance structures among local governments, civic organizations, educational institutions, and digital-service providers.

2.2.2 Policy Distinctions: People-Based, Place-Based, and Place-Sensitive Approaches

The two types of factors reviewed in the preceding section correspond to the distinction in public policy design between a people-based approach and a place-based approach. People-based policies seek to improve outcomes by directly supporting human characteristics such as individual competency, income, and education. Place-based policies seek to indirectly improve outcomes for residents of a given area by improving that area's infrastructure, institutions, and environmental conditions (Neumark & Simpson, 2015). The two approaches are not mutually exclusive, but because the allocation of policy resources and the pathways of intervention differ, they need to be designed separately from the perspective of policy implementation.

However, the relationship between the two approaches is not a simple parallel. Hindman's (2000) time-series analysis of the United States showed that income, age, and education are stronger predictors than geography, while also reporting a pattern in which disparities become entrenched over time. This implies that individual characteristics and regional structures can interact over time. van Dijk (2020) proposed a sequential model moving from lack of motivation to material access, skills, and substantive use, suggesting that individual factors and structural factors operate cumulatively.

In this context, recent policy and academic literature has been moving away from an either-or choice between people-based and place-based approaches and toward a place-sensitive approach that understands people's experiences within the context of place. One example is the attempt to recognize lived experience and place-based insight as forms of knowledge equivalent to quantitative evidence (OECD, 2025). This approach requires that people-based factors and place-based factors be analytically separated, while also considering how the two combine in specific places.

2.2.3 Metatheoretical Framework: Critical Realism's Stratified Ontology

This project adopts the basic principle of its analytical design, and grounds the metatheoretical foundation of that design in the stratified ontology of critical realism. According to Bhaskar (1975), the world consists of three domains. The real domain includes structures, mechanisms, and causal powers that exist independently of whether they are observed. The actual domain is the domain in which these structures and mechanisms operate under particular conditions and generate events. The empirical domain refers to the subset of events that have occurred and are observable. Not all events in the actual domain are experienced, nor do observed phenomena represent the entirety of the real domain.

This ontological distinction entails the logic of retroduction (Bhaskar, 1975). Retroduction is a method that infers backward from observed empirical events to the structures and mechanisms that made those events possible. It is distinct from induction and deduction, and asks, "What structure must exist for this phenomenon to be possible?"

2.2.4 Bayesian Updating and Evidence Integration

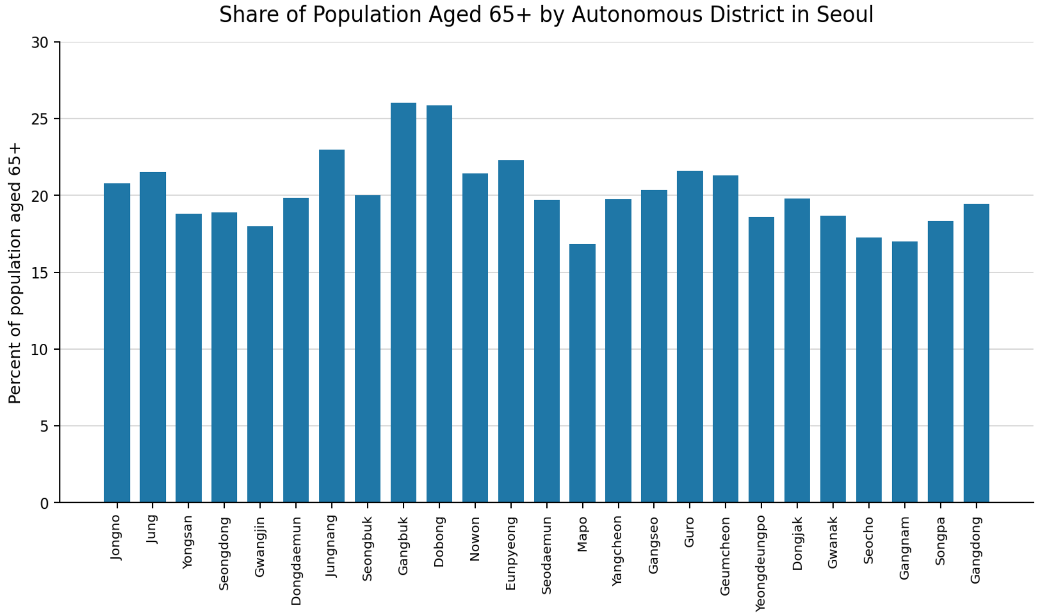
Bayesian updating provides a transparent way to combine existing evidence with new evidence. In ordinary terms, it starts with a prior belief based on the information already available, observes new data, and then produces an updated belief, called a posterior. The method is useful when evidence is uneven across places because it does not treat every new observation as equally decisive. Instead, the strength of the update depends on both the prior information and the amount and reliability of the new evidence (Gelman et al., 2013; Kruschke, 2014).

2.3 Seoul

2.3.1 Seoul as a City

Seoul, the capital of the Republic of Korea, is a highly dense metropolis in which 20% of the national population resides on 0.6% of the country's total land area. Because of its high density and concentration of technological capacity, Seoul has stood at the forefront of South Korea's digital transformation. Yet a closer look inside Seoul reveals meaningful disparities associated with the demographic structure and economic self-sufficiency of each of Seoul's 25 districts. This section examines the characteristics of each district in Seoul and discusses the features of Seoul's population geography.

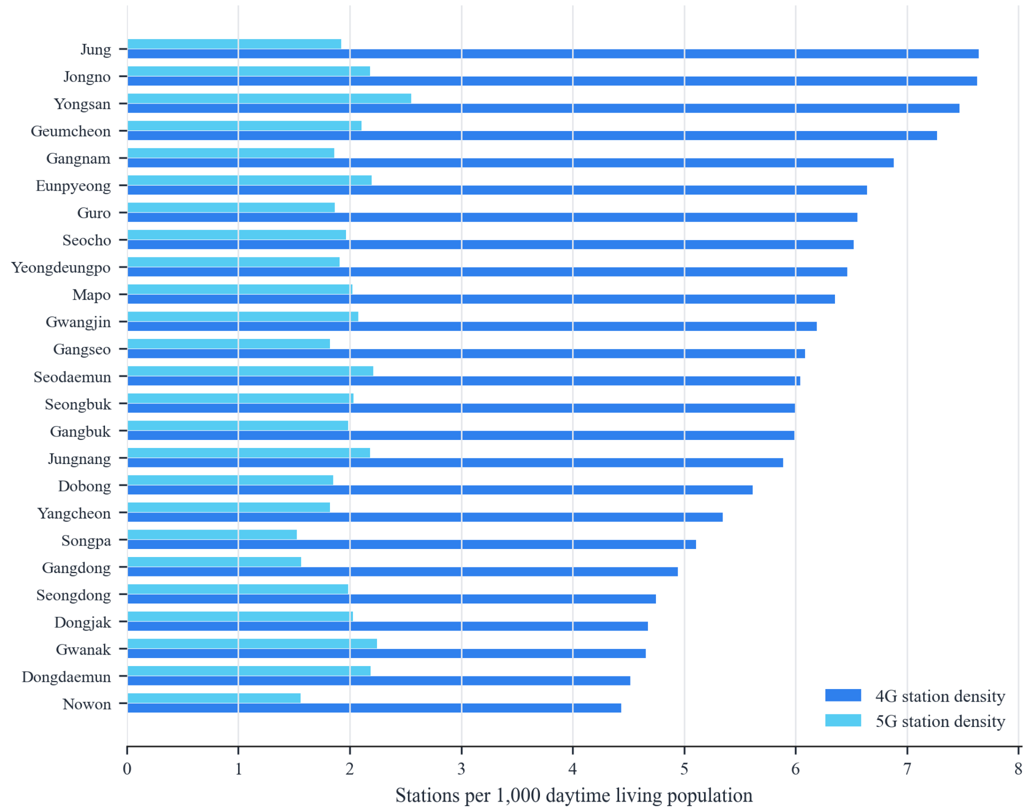
In recent years, Seoul's population geography has been reorganized along two axes: rapid aging and the expansion of single-person households. Northern peripheral districts such as Gangbuk-gu(26.03%) and Dobong-gu(25.85%) already have shares of older adults exceeding 25% and exhibit characteristics of a super-aged society, whereas the Gangnam area and neighborhoods around major universities maintain relatively younger population structures, making inter-district imbalances in age distribution pronounced. These demographic differences, combined with each district's average income level and economic vitality, form the structural backdrop that produces the differences in Affordability and Devices capabilities identified in Section 3.1 of this study.



In addition, Seoul's policy implementation system operates through organic cooperation between the metropolitan government, the Seoul Metropolitan Government, and the 25 lower-level districts. The broad direction of policy follows a top-down approach established by Seoul City Hall, which plays a central role in building metropolitan-scale infrastructure and setting standard guidelines for digital inclusion. At the same time, Seoul also operates bottom-up decision-making structures, including the citizen participatory budgeting system and residents' associations, through which digital inconveniences experienced by local residents in everyday life are directly reflected in policy. Based on these metropolitan plans, region finalize budgets through local council deliberation and ordinance enactment, and ultimately deliver close-to-resident services to citizens through neighborhood community service centers. In this sense, Seoul's policies for reducing the digital divide can be understood as being realized through a multilayered process that combines the administrative capacity of each district with citizen participation, rather than through uniform central execution.

2.3.2 Current Status of Connectivity Gaps and Related Laws and Policies

The current state of digital connectivity in Seoul clearly reveals the limits of privately led infrastructure deployment through disparities in radio station infrastructure density across districts. According to Korea Radio Promotion Association statistics on radio stations for 2020-2025, the core infrastructure determining Seoul's communication quality, namely base stations and mobile repeaters, is overwhelmingly concentrated in particular areas where profitability is high and traffic is concentrated. Southeastern districts such as Gangnam-gu(base stations 19,190; mobile repeaters 2,711) and Songpa-gu(base stations 14,751) possess dense networks, whereas northern peripheral districts such as Dobong-gu(base stations 4,008) and Gangbuk-gu(base stations 4,700) show a gap of up to 4.8 times in the number of base stations. In particular, for simple radio stations used for small businesses and Internet of Things(IoT) connectivity, Gangnam-gu(24,236) has about 20 times as many as Dobong-gu(1,224), suggesting that a rich-get-richer and poor-get-poorer pattern in the digital ecosystem across districts has already become entrenched at the level of physical infrastructure.



To correct these regional imbalances in private infrastructure and guarantee citizens' basic right to communication, Seoul made the policy decision to step forward directly as a facilities-based telecommunications service provider as an administrative alternative. Under the 2025 amendment to the Telecommunications Business Act, Seoul became the first local government to secure the status of a telecommunications service provider, and it is expanding public Wi-Fi and IoT infrastructure by directly connecting more than 5,000 km of its own fiber-optic cable network to underserved areas where private telecommunications companies have invested insufficiently because of profitability concerns. Seoul currently operates more than 34,000 public Wi-Fi units, the largest scale in the country; these are widely installed on streets, in parks, markets, welfare facilities, and city buses, generating an annual reduction of approximately KRW 200 billion in citizens' communication costs. In particular, mobile Wi-Fi installed on roughly 7,000 city buses has become a form of digital infrastructure that citizens experience most directly in everyday life. Through the expansion of public Wi-Fi, Seoul plans to reduce disparities in information access and ease the household burden of communication costs amounting to hundreds of billions of won each year. It also seeks to establish a sustainable virtuous cycle by lowering dependence on privately leased networks and reinvesting the secured resources in projects that benefit citizens.

Policy responses to address these infrastructure gaps have moved beyond physical network expansion and increasingly focus on strengthening Affordability by lowering citizens' actual cost burden. The central and local governments operate the Smart Choice platform to help users understand their own communication consumption patterns and make rational choices. Smart Choice provides integrated services such as checking handset subsidies, recommending customized rate plans, and searching for unclaimed refunds, thereby reducing information asymmetry and encouraging reductions in household communication costs.

The recent reform of 5G rate plans is especially significant in that it shifted a market structure centered on high-priced plans toward one centered on low- and mid-priced plans. The lowest 5G price tier, which had previously remained in the KRW 40,000-50,000 range, was lowered to the KRW 30,000 range, and intermediate plan tiers between 30 and 100GB were newly and more finely introduced to correspond to actual data usage. This reform addressed the structural contradiction in which users paid high costs for data they did not use, and by 2024 it had led to a concrete policy outcome in which more than 6 million subscribers switched to low- and mid-priced plans.

Customized dedicated rate plans for vulnerable groups and specific age groups have also been substantially strengthened. For young people(age 34 and below), the three major mobile carriers in the Republic of Korea expanded the amount of data provided by up to twice that of ordinary plans at the same price, taking into account high data demand among this group, and broadened the eligible population by raising the upper age limit for subscription from age 29 to 34. In addition, for older adults(age 65 and above), LG U+ launched dedicated rate plans segmented by age group(65, 70, 80, and so on) that are up to 20% cheaper than ordinary plans, reflecting older adults' relatively lower data usage and higher price sensitivity. Furthermore, in line with the trend toward non-face-to-face subscription, online-only rate plans have expanded choice by introducing no-contract plans that are approximately 30% cheaper than ordinary plans when users subscribe through online channels without visiting a retail store.

As a result, Seoul is building a multifaceted inclusion policy that guarantees universal digital basic rights for all citizens through a three-part framework: information transparency through Smart Choice, the rational rate structures of the three major mobile carriers, and Seoul's direct provision of public infrastructure.

In addition, the most significant change in Seoul's administrative services in 2026 is the unification and upgrading of mobile platforms. Seoul will terminate the previously fragmented Seoul Citizen Card and Seoul Wallet services as of March 31, 2026, and consolidate their functions into the integrated mobile platform Seoul ON. By enabling citizens to address everything from the use of public facilities such as libraries and sports facilities to verification of disability or national merit status within a single app, this change lowers the entry barrier to digital administration created by service fragmentation. In particular, Seoul ON enhances administrative efficiency by strengthening the function that verifies eligibility for vulnerable-group status in real time through linkage with public MyData, without requiring separate document submission.

Furthermore, Seoul is pursuing an integrated inclusion policy centered on the AI Digital Learning Center so that infrastructure expansion and service integration lead to genuine improvements in citizens' practical usage capabilities. Using nationwide digital learning center sites, including 10 sites in Seoul, the city provides ongoing face-to-face education for older adults and digitally vulnerable groups, and offers not only basic internet-use instruction but also life-oriented AI experience education and education to prevent digital dysfunction. In particular, it is closing educational blind spots by expanding "visiting 1:1 education" for people such as those with severe disabilities who have mobility difficulties, and by establishing a level-specific customized education system linked across online and offline formats. This approach reflects the judgment that even when hardware(infrastructure) and software(administrative services) are in place, gaps will not be resolved if people lack the human capabilities needed to use them. In this way, Seoul is substantively realizing Universal Meaningful Connectivity(UMC) through a three-part framework: securing infrastructure stability based on law, innovating the service delivery system through Seoul ON, and improving users' capabilities through education.

3. Analysis

The three analyses in this chapter are sequentially connected. First, the study constructs a UMC index by district in Seoul to measure relative disparities (3.1). It then uses these results as district-level variables to analyze associations between individual and district characteristics and individuals' digital connectivity (3.2). Finally, it identifies platform-visible cases that are not captured by quantitative analysis through digital platform text. This sequence consists of measurement, explanatory decomposition, and exploratory lived-experience signal detection. It is designed to connect district-level conditions, individual-level vulnerabilities, and everyday narratives without reducing one level to another.

3.1 Regional Disparities in UMC in Seoul

3.1.1 Measurement Indicators and Data

The measurement indicators constructed in this project are based on the conceptual definitions of the six dimensions of UMC. These dimensions are classified as Connectivity, Available for Use, Affordability, Devices, Digital Skills, and Safety, and the individual measurement indicators were specified according to two core components that simultaneously consider meaningfulness and universality (ITU, 2023). The measurement indicators for the six dimensions comprising Seoul's UMC used in this analysis are shown in Table 1. In selecting measurement indicators, the analysis comprehensively considered alignment with conceptual definitions, Seoul's digital technology environment, and data availability. The analysis used data from two time points, 2023 and 2024, the most recent years for which data could be secured. National administrative data and related sources from these two time points were used. However, because of limitations in the timing of data provision, the Devices, Digital Skills, and Safety dimensions applied the same values for both time points based on 2023 survey data.

Connectivity, measurement indicators were calculated using mobile radio station data provided by Korea Radio Promotion Association and living population data from Seoul's administrative data. According to ITU report (2023), Connectivity can be measured based on the population residing within network coverage. In Korea, however, the separation between workplace and residence is often pronounced, so the area of residence may differ from the area where individuals actually access the digital environment. Accordingly, this study devised an indicator more appropriate to the Korean context by calculating 4G and 5G network coverage using daytime floating population data rather than simple resident population. Specifically, the numbers of 4G base stations and 5G base stations were each converted into the number of base stations per 1,000 daytime living population at the enumeration-district level, and the final indicator used the average of these values by district.

Available for Use is the dimension that measures the practical usability of digital services. Monthly mobile data usage per household within each area, provided by SKT communication information, and online service usage days in four categories, finance, shopping, video, and delivery, were aggregated monthly by district as population-weighted averages, and the 12-month average values were used. In addition, data on public Wi-Fi registration status from district governments were used to calculate and include the number of public Wi-Fi APs per 1,000 total population. Public Wi-Fi density functions as an especially important digital access pathway for older adults whose home internet access environment is limited (Van Dijk, 2020).

Affordability was constructed following the two-axis structure of the ITU ICT Development Index 2023(ITU, 2024). For the purchasing power(capacity) side, the analysis used average monthly household income by district provided by Seoul's Commercial District Analysis Service. For the price burden(burden) side, it used the share of bill arrears within the most recent three months from SKT communication information. The arrears rate was calculated as a population-weighted average by administrative dong, sex, and age group, and then aggregated to the district level; it is a reverse-coded indicator, meaning that higher values indicate lower economic accessibility.

Devices measures the possession and diversity of use of digital devices. The weighted averages by district were calculated for the number of core digital device types owned within households and the number of device types actually used by individuals, based on the Seoul Citizens' Digital Competency Survey. According to ITU (2023), possession and use are interpreted differently depending on device characteristics, such as mobile phones and computers. In the Korean domestic context, whether a computer is owned can also be an indicator of digital connectivity. In Seoul, where smartphone penetration is effectively saturated, ownership of auxiliary devices such as desktops, laptops, and tablets acts as a differentiating factor in digital usage capability(Van Deursen & Helsper, 2015).

Digital Skills comprises four subdomains: Basic, Applied, Advanced, and Cyber activity. Basic competency measures the basic ability to operate smart devices; Applied competency measures the ability to use digital services; Advanced competency measures proficiency in smart-device activities; and Cyber activity competency measures the ability to act in cyberspace. The raw score for each subdomain was divided by its maximum possible score and converted to a 0-1 ratio, after which weighted averages were calculated by district.

Safety comprises two subindicators: digital security behavior and security awareness. Security behavior measures actual security practices, such as password management, software updates, and use of two-factor authentication, while security awareness measures the level of recognition of the importance of personal information protection. Each subindicator was converted to a ratio relative to its maximum possible score, and weighted averages were then calculated by district.

Table 1. UMC Dimension Indicators

Dimension	Indicator	Data Source	Note
Connectivity	station_density_4g	Spectrum Resource Mgmt System	Per 1,000 daytime living population
	station_density_5g	Spectrum Resource Mgmt System	Per 1,000 daytime living population
	download_speed	NIA Communication Quality	2022–2024 average
Available for Use	mobile_data_usage	SKT Telecom Data	Population-weighted district average
	online_service_days	SKT Telecom Data	Population-weighted district average
	wifi_density	Local Administrative Wi-Fi License Data	Per 1,000 total population
Affordability	avg_income	Seoul Commercial Analysis Service	ITU IDI capacity dimension
	delinq_rate	SKT Telecom Data	Reverse-coded; ITU IDI burden dimension
Devices	device_home_core	Seoul Citizens' Digital Competency Survey 2023	Excluding smartphones
	device_use_core	Seoul Citizens' Digital Competency Survey 2023	Including smartphones
Digital Skills	Basic	Seoul Citizens' Digital Competency Survey 2023	—
	Applied	Seoul Citizens' Digital Competency Survey 2023	—
	Advanced	Seoul Citizens' Digital Competency Survey 2023	—
	Cyber activity	Seoul Citizens' Digital Competency Survey 2023	—
Safety	safety_behavior	Seoul Citizens' Digital Competency Survey 2023	Security practices (7 items)
	safety_awareness	Seoul Citizens' Digital Competency Survey 2023	Awareness (3 items)

3.1.2 Indexing the Measurement Indicators

Following the theoretical discussion in Section 2.1, Min-Max scaling is used here because Seoul is a high-connectivity city in which many basic UMC thresholds have already been met. The original UMC logic is threshold-based, but in this case a simple threshold test would not reveal the remaining differences among the 25 districts. The indicators are therefore normalized to the [0, 1] interval through within-Seoul Min-Max scaling so that the analysis can compare each district's relative position after heterogeneous indicators are placed on a common scale. The normalized value x* is calculated separately for each indicator and year as follows.

$$\text{Positive indicator: } x^* = \frac{x - min(x)}{max(x) - min(x)}$$

$$\text{Reverse indicator: } x^* = \frac{max(x) - x}{max(x) - min(x)}$$

Reverse-direction indicators, for which larger values indicate worse outcomes, were transformed so that higher normalized values consistently represent more favorable UMC conditions. Specifically, the telecom fee delinquency rate (delinq_rate) was reverse-coded. Within each of the six UMC dimensions - Connectivity (score_Infrastructure), Available for Use, Affordability, Devices, Digital Skills, and Safety - the normalized indicators were combined by an equally weighted arithmetic mean to produce dimension scores, and the final score_UMC was calculated as the equally weighted mean of the six dimension scores. Equal weights were used to minimize arbitrary researcher judgment in the absence of an agreed standard for the relative importance of the six dimensions (ITU, 2024). The underlying data follow a fixed pipeline: data/raw is the protected source layer, data/processed holds cleaned district-level inputs, and report-grade outputs are stored in output/tables and output/figures. The resulting composite index ranges from 0 to 1, was calculated independently for 2023 and 2024, and should be interpreted as each district's relative position within Seoul rather than as an absolute UMC attainment level.

3.1.3 Characteristics of the UMC Index by District

> **KEY FINDING**
> Seoul's measured UMC gap is multidimensional and district-specific. The 2024 composite score ranges from 0.279 in Jungnang-gu to 0.695 in Seocho-gu, and the provisional Digital Desert group (Jungnang-gu, Dobong-gu, Gangbuk-gu, Guro-gu, and Nowon-gu) is weakest in Devices, Safety, and Affordability rather than simple physical connectivity.

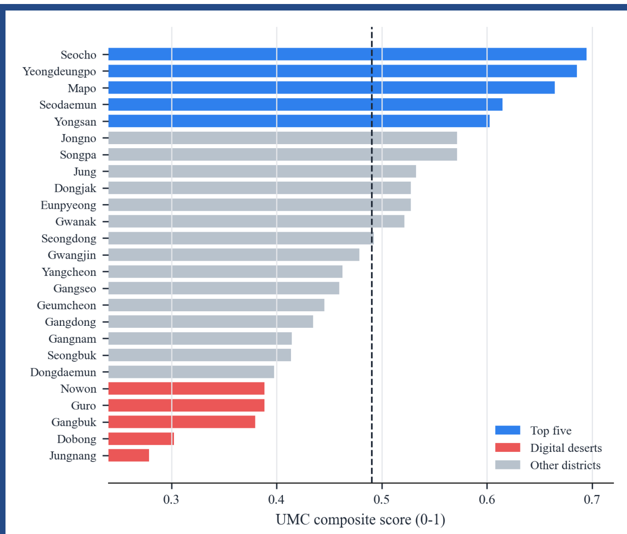
The 2024 UMC composite index for the 25 districts ranged from 0.279 (Jungnang-gu) to 0.695 (Seocho-gu), indicating an approximately 2.5-fold gap between the highest and lowest districts. Despite the common perception that Seoul is a city with high internet penetration, a substantial relative digital connectivity divide was observed at the lower administrative-unit level. The top five districts were Seocho-gu (0.695), Yeongdeungpo-gu (0.686), Mapo-gu (0.665), Seodaemun-gu (0.615), and Yongsan-gu (0.603). The bottom five were Jungnang-gu (0.279), Dobong-gu (0.303), Gangbuk-gu (0.380), Guro-gu (0.389), and Nowon-gu (0.389). Higher-ranked districts were generally located in the central city or the Gangnam area, benefiting from dense commercial activity, high income levels, and abundant private digital infrastructure, whereas lower-ranked districts were concentrated in the northern periphery and the southwest. The 25-district mean of the 2024 composite score was 0.4905, and the six dimension means were relatively close to one another—Connectivity 0.512, Available for Use 0.490, Affordability 0.469, Devices 0.468, Digital Skills 0.500, and Safety 0.505—indicating that the overall city-level gap is shaped less by a single dominant dimension than by the way district-level strengths and weaknesses are distributed across dimensions. The dimension-specific extremes confirm this heterogeneity: Connectivity peaked in Yongsan-gu (0.884) against a low of Nowon-gu (0.174); Available for Use was led by Gwangjin-gu (0.832) with Dobong-gu (0.213) at the bottom; Affordability ranged from Seocho-gu (0.993) to Gangbuk-gu (0.000); Devices ranged from Seocho-gu (0.988) to Guro-gu (0.033); Digital Skills topped at Gwanak-gu (0.795) with Jungnang-gu (0.242) lowest; and Safety reached its maximum in Jongno-gu (0.984) with Dobong-gu (0.019) at the floor.

Examining the spatial distribution of the composite index (Figure 4), districts with high UMC index values were concentrated in two areas. These were the Han River and central-city corridor running from Seocho to Yeongdeungpo and Mapo, and the traditional downtown area centered on Yongsan, Jongno, and Jung-gu. By contrast, the northern periphery (Gangbuk-gu, Dobong-gu, Nowon-gu) and the southwest (Guro-gu, Geumcheon-gu) scored below the city average. This spatial distribution was associated with the distribution of socioeconomic resources in Seoul. In other words, it suggests that the digital divide may be closely linked to structural spatial inequality in the city.

At the same time, the dimension-level analysis found that no district performed uniformly well across all six dimensions (Figure 3). For example, Eunpyeong-gu in northern Seoul ranked high in the Connectivity dimension, with a score of 0.781, but remained in the lower tier in the Affordability dimension, with a score of 0.328. Similarly, Gwanak-gu ranked first in the Digital Skills dimension, with a score of 0.795, but fell into the lowest tier for Affordability, at 0.152. These multidimensional inter-district inequalities indicate that policy-relevant dimension-specific vulnerabilities cannot be captured by the composite ranking alone.

In addition, the composite UMC index shows statistically significant spatial clustering, while the dimension-level LISA panels identify selected local clusters rather than uniform citywide clustering in every dimension. Figure 5 therefore supports a cautious interpretation: spatially patterned district conditions can justify metropolitan coordination for provisional Digital Deserts, but dimension-specific policy design should remain tied to the measured profile of each district.

In conclusion, as shown in Figures 4 and 5, this project provisionally classifies the five districts in the lowest group of the composite index (Jungnang-gu, Dobong-gu, Gangbuk-gu, Guro-gu, and Nowon-gu) as Digital Deserts. These districts commonly exhibit clear deficits in the Devices and Safety dimensions, and Affordability also falls below the Seoul average. Meanwhile, in the Connectivity dimension, some districts (Jungnang-gu 0.630, Gangbuk-gu 0.562) exceed the Seoul average, indicating that the core divide may stem from insufficient human and economic capabilities to use physical network infrastructure rather than from the absence of that infrastructure itself. This finding is examined more systematically in Section 3.2 by separating and testing the relative contributions of individual-level and district-level variables in a multilevel analysis. To preserve a consistent measurement basis across years and to mitigate endogeneity, the Level 2 covariates carried into Section 3.2 are drawn primarily from the infrastructure-side dimensions (Connectivity, Available for Use, and Affordability), whereas the Devices, Digital Skills, and Safety dimensions - largely constructed from the 2023 Seoul Survey responses that also enter the Level 1 outcome - are intentionally held out of the main multilevel specification to avoid same-source contamination.



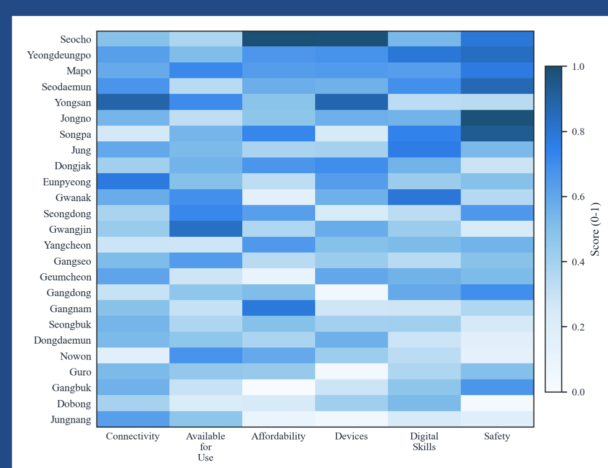


Figure 2. UMC Composite Scores by District, 2024

Figure 3. Heatmap of UMC Dimension Scores, 2024

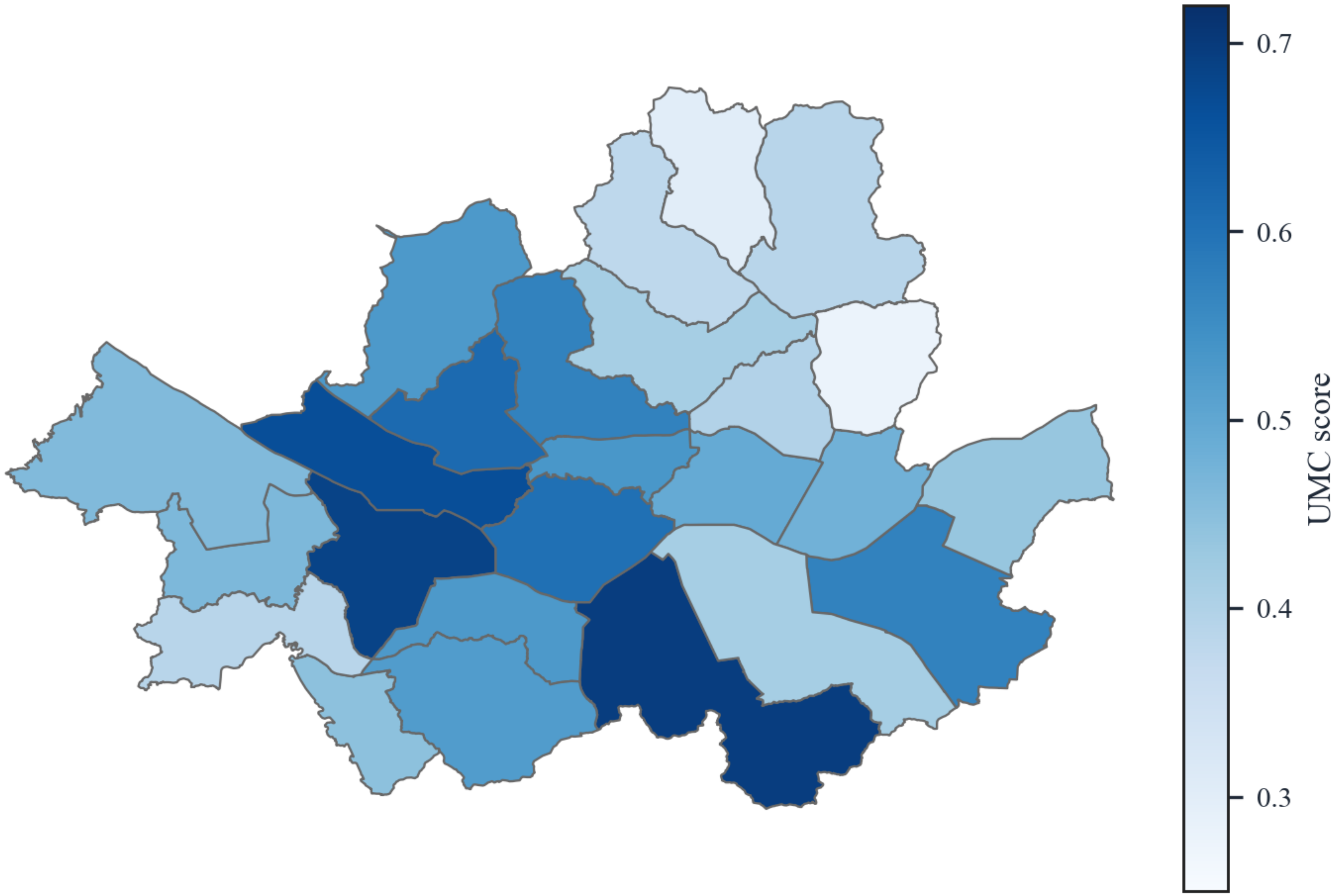


Figure 4. Spatial Distribution of the UMC Composite Index, 2024

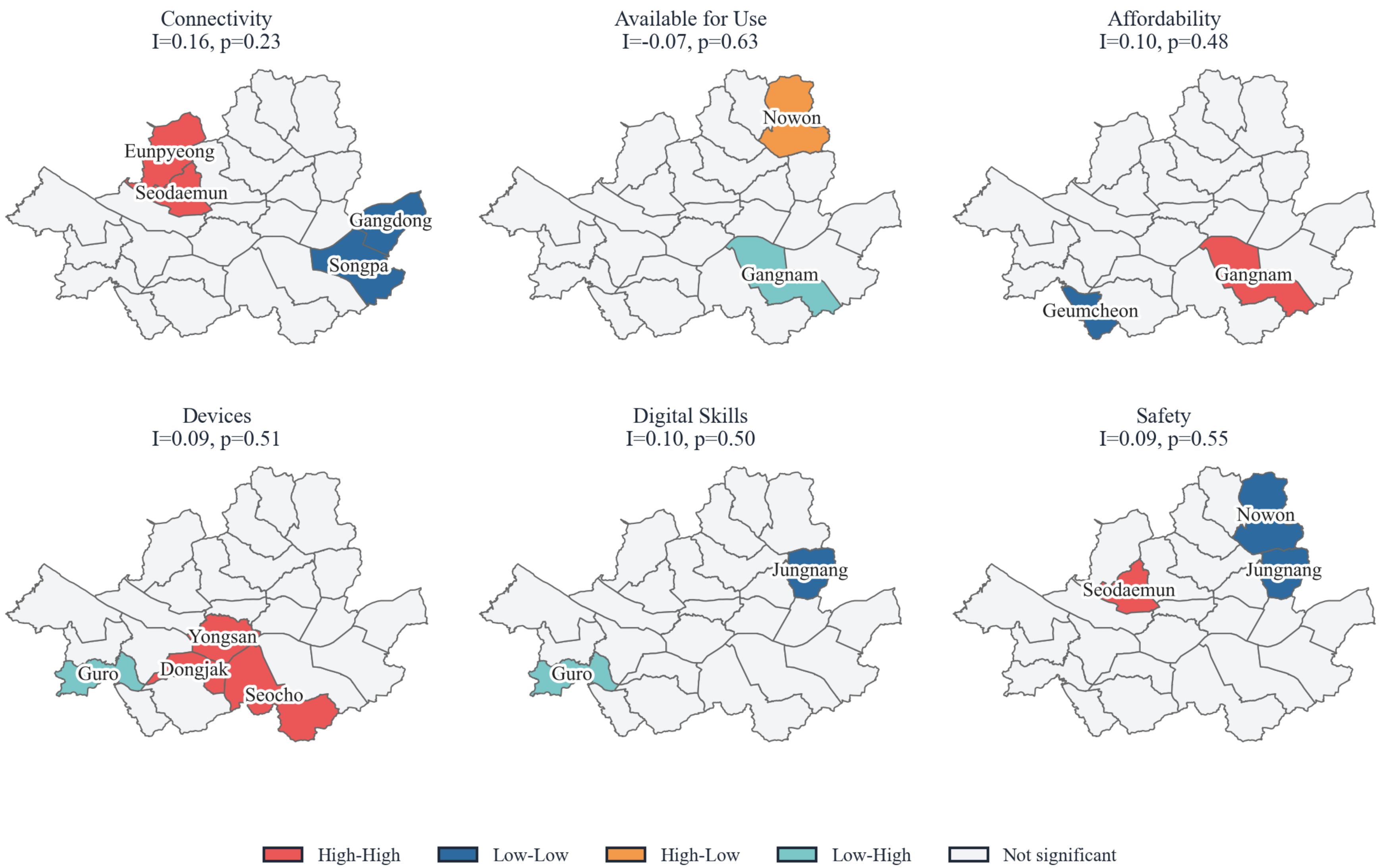


Figure 5. Spatial Autocorrelation and LISA Clusters by UMC Dimension, 2024

3.2 Individual and District Characteristics and Digital Connectivity

The data used in this study have a hierarchical structure in which individuals (Level 1) are clustered within districts (Level 2). District-year UMC scores are merged to respondents by residential district and survey year, while random intercepts are estimated for Seoul's 25 districts. In this structure, conventional OLS regression can violate the assumption of independent observations, underestimate the standard errors of district-level predictors, and inflate Type I error (Raudenbush & Bryk, 2002). Therefore, this study adopted a two-level hierarchical linear model (HLM) that decomposes variance into within-district and between-district components and allows individual-level and district-level variables to be entered simultaneously.

The Seoul Survey, used in the second analysis, is administrative data that the Seoul Metropolitan Government has collected annually since 2003 from 20,000 households. Data collection is conducted every August-September through a combination of non-face-to-face and in-person interviews.

3.2.1 Variable Construction

First, the dependent variable in the analysis, the digital usage score, is the summed score of eight items on digital service usage frequency among the Seoul Survey response items. The eight items consist of living information services, e-commerce services, financial transaction services, public services, communication services, leisure services, education and learning services, and mobile payment. These are not basic acts of connection such as phone calls or text messaging, but rather substantive additional uses mediated by digital devices, and were therefore judged to be indicators suited to Seoul's advanced digital environment. Each item was transformed from its original four-point ordinal scale into a 0-10 continuous scale, yielding a continuous variable ranging from 0 to 80 points. A higher score means that an individual uses a more diverse range of digital services and can freely use desired services.

The independent variables in this analysis are divided into individual-characteristic variables and district-characteristic variables (Table 2). The individual-characteristic variables consist of sex, age, occupation, educational attainment, income level, and disability status, all of which may affect individual digital usage. Educational attainment was divided into four categories: middle school completion or below, high school completion, college graduation (reference category), and graduate school or above. Occupation was divided into four categories: managers and professionals, clerical workers (reference category), production and technical workers, and other (students, homemakers, and unemployed persons). Housing type was classified into four categories: apartment (reference category), detached house, multiplex and multi-household housing, and other. Age and income were grand-mean centered to improve the interpretability of the intercept. In addition, a single-person household dummy, a disability-status dummy, and a year dummy (reference: 2023) were included.

As Level 2 variables, the analysis used three UMC dimension scores constructed in Section 3.1 that describe local digital contextual conditions: Connectivity, Available for Use, and Affordability. This choice reflects that Devices, Digital Skills, and Safety are aggregated to the district level from individual responses in the Seoul Citizens' Digital Competency Survey. To avoid the resulting endogeneity problem, these three dimensions were excluded from the main analysis. All Level 2 UMC variables were standardized to the [0, 1] range using Min-Max normalization and merged by district and survey year.

The analysis proceeds in four steps. After estimating an unconditional model and calculating the intraclass correlation coefficient (ICC) to support the use of multilevel analysis, the study sequentially estimates M1, which includes individual-level variables; M2, which adds the three district-level UMC dimensions; and M3, which includes cross-level interactions and key within-level interactions. The cross-level interactions are constructed as products of UMC contextual dimensions and vulnerable-group indicators, while the within-level interactions test compound vulnerability among older adults living alone and respondents with lower educational attainment as age increases.

3.2.2 Analysis Results

KEY FINDING
Individual vulnerability is the dominant structure in the HLM results. The ICC is only 0.49%, while lower education, age, older-adult single-person households, and the age x lower-education pattern remain more consequential than district-level UMC dimensions. District indicators are therefore targeting context, not causal proof of infrastructure effects.

The sample used in the analysis consists of a total of 10,000 Seoul Survey respondents from 2023-2024, with an average of 400 sampled from each of the 25 districts. Table 3 presents descriptive statistics for the main variables. The mean value of individual digital connectivity was 60.88 out of 80 points, demonstrating the high level of digital connectivity among Seoul residents. Although the dependent variable may exhibit negative skewness, the analysis was conducted using the scale-linearized raw scores, given the institutional standard scale and a sample size approaching 10,000.

The sequential estimation results of the multilevel models are presented below. The intraclass correlation coefficient (ICC) of the unconditional model was 0.49%, indicating that the share of between-district variance itself was not large. However, the design effect, calculated in light of the average sample size of 400 persons per district, was 2.94, exceeding the threshold of 2.0 and supporting the use of a multilevel model (Hox et al., 2018). Nevertheless, taken together, the small number of 25 districts and the very small share of variance explained by district suggest that district-level connectivity-related characteristics have a limited average association with individual digital connectivity. These results may indicate that, because Seoul has excellent digital connectivity and a high level of digital competency, district-level connectivity infrastructure does not show a large average association. In addition, because the Seoul Survey records individuals' residential districts, those districts may not correspond to the areas where individuals are directly affected by connectivity.

Among individual characteristics, the association with educational attainment was the most pronounced. In Models 1 and 2, education of middle school completion or below was associated with a digital usage score about 22.6 points lower than that of college graduates; even in the interaction model, the gap remained about 19.0 points. High school graduates also scored about 7.3 points lower than college graduates, whereas those with graduate school education or above scored about 2.9 points higher, indicating a clear positive relationship between educational attainment and digital usage capacity. Age also showed a strong negative association. For each one-year increase from the mean age, the score for individual digital connectivity decreased by about 0.35 points in Models 1 and 2 and by 0.28 points in the interaction model. Among older adults aged 65 or above, an additional 6.0-point decline was observed apart from the age association.

Table 2. Variable Descriptions

Variable	Description	Level	Scale	Source
digital_use_score	Digital service usage score (sum of 8 items)	L1	0–80 (continuous)	Seoul Survey 2023–24
age_c	Age (grand-mean centered, M = 49.3)	L1	years	Seoul Survey
female	Female (ref: male)	L1	0/1	Seoul Survey
Edu	Education: ≤ middle school / high school / graduate school or higher (ref: college)	L1	0/1 each	Seoul Survey
Income	Monthly income (grand-mean centered)	L1	10K KRW	Seoul Survey
job	Job: professional / blue-collar / other (ref: white-collar)	L1	0/1 each	Seoul Survey
disabled_bin	Disability status	L1	0/1	Seoul Survey
hh_size_1	Single-person household	L1	0/1	Seoul Survey
housing_detached / _multi / _other	Housing: detached / multi-family / other (ref: apartment)	L1	0/1 each	Seoul Survey
year	Year 2024 dummy (ref: 2023)	L1	0/1	Seoul Survey
elderly	Older adults (≥65)	L1	0/1	Seoul Survey
low_edu	Lower educational attainment (≤ middle school)	L1	0/1	Seoul Survey
Connectivity	UMC connectivity/infrastructure dimension score (normalized)	L2	0-1	UMC Part 1
Available for Use	UMC available-for-use dimension score (normalized)	L2	0-1	UMC Part 1
Affordability	UMC affordability dimension score (normalized)	L2	0-1	UMC Part 1

In terms of occupation, managers and professionals scored about 2.3 points higher than clerical workers. Residents of detached houses scored about 2.5 points lower than apartment residents, and residents of multiplex or multi-household housing scored about 1.8 to 1.9 points lower. These associations suggest that housing type may reflect differences in income, residential environment, or in-home connectivity conditions. Single-person households scored about 1.8 points lower in Models 1 and 2, although the main effect became smaller once the older-adult single-person interaction was introduced. Among other individual characteristics, women scored about 2.6 points higher than men, and respondents with disabilities scored about 3.0 points higher. The positive coefficient for disability status should be interpreted cautiously as a sample-specific association rather than as evidence of a general advantage.

At the district level, the UMC dimensions do not show a uniform positive average effect. In Model 2, Connectivity and Available for Use are not statistically significant, while Affordability is marginally negative. In Model 3, Connectivity and Affordability have statistically significant negative coefficients, whereas Available for Use remains non-significant. Given the very small between-district variance, these district-level coefficients should be interpreted cautiously as contextual associations rather than as evidence that infrastructure worsens digital usage.

The cross-level interaction results are more limited than the earlier draft suggested. The interaction between Available for Use and older adults is positive but not statistically significant, and the Connectivity x lower educational attainment interaction is also not statistically significant in the current main model. The only cross-level interaction that approaches conventional significance is Affordability x older adults, which is marginally positive. Therefore, the current results do not support a strong claim that public Wi-Fi density or connectivity infrastructure has a clearly significant differential effect for older adults or lower-education residents.

At the individual level, the interactions indicate a pattern of double vulnerability. Older adults living alone scored an additional 4.0 points lower than older adults not living alone, and the negative association between lower educational attainment and digital use became larger with age. These findings show that compound vulnerability factors intensify the digital divide more than any single vulnerability factor.

Taken together, the HLM results identify the dominant structure of digital usage inequality in Seoul: individual-level vulnerabilities dominate digital usage disparities, while district-level indicators operate mainly as contextual conditions. Educational attainment, age, older-adult status, single-person household status, and the interaction between lower education and aging show stronger and more consistent associations than the district-level UMC dimensions. District scores therefore remain useful for identifying where vulnerable individuals may be concentrated or easier to reach, but they should not be interpreted as proving that district infrastructure itself causes individual digital usage inequality.

Table 3. Descriptive Statistics

Variable	N	Mean	SD	Min	Max
Digital usage score	10,000	60.88	16.84	0	80
Age	10,000	49.32	16.23	15	99
Female, proportion	10,000	0.51	–	0	1
Education: ≤ middle school	10,000	0.07	–	0	1
Education: high school	10,000	0.25	–	0	1
Education: graduate school or higher	10,000	0.13	–	0	1
Disability status	10,000	0.03	–	0	1
Monthly income (10K KRW)	10,000	237.55	209.73	0	1,050
Single-person household	10,000	0.17	–	0	1
Older adults (age 65+)	10,000	0.22	–	0	1
Cluster-level Connectivity Index (Connectivity, norm.)	50	0.5	0.18	0.18	0.98
Available for Use (norm.)	50	0.5	0.19	0.12	0.96
Affordability (norm.)	50	0.5	0.22	0.07	1

Note. N = 10,000 individuals nested within j = 25 districts (2023-2024 pooled). District-level UMC variables have 50 district-year observations because 2023 and 2024 UMC scores are merged by district and survey year. Income is measured in 10,000 KRW units. District-level variables are min-max normalized indices (0-1 scale).

Table 4. HLM Estimation Results

Variable	Model 1	Model 2	Model 3
Intercept	63.204*** (0.416)	64.282*** (0.764)	65.283*** (0.811)
Level 1: Individual characteristics			
Age (centered)	-0.347*** (0.012)	-0.347*** (0.012)	-0.278*** (0.014)
Female	2.636*** (0.305)	2.659*** (0.305)	2.602*** (0.305)
Education: middle school or below	-22.618*** (0.770)	-22.674*** (0.770)	-19.023*** (2.379)
Education: high school	-7.289*** (0.380)	-7.330*** (0.381)	-7.261*** (0.380)
Education: graduate school or higher	2.906*** (0.483)	2.952*** (0.483)	2.915*** (0.481)
Disability	3.084*** (0.888)	3.040*** (0.888)	3.008*** (0.884)
Income (centered)	0.003*** (0.001)	0.003*** (0.001)	0.002** (0.001)
Job: professional	2.329*** (0.451)	2.331*** (0.451)	2.280*** (0.449)
Job: blue-collar	-0.373 (0.424)	-0.378 (0.424)	-0.576 (0.423)
Job: other	-1.233** (0.456)	-1.213** (0.456)	-1.102* (0.454)
Single-person household	-1.775*** (0.338)	-1.789*** (0.338)	-0.874* (0.367)
Detached house	-2.478*** (0.377)	-2.486*** (0.379)	-2.567*** (0.378)
Multi-family housing	-1.798*** (0.408)	-1.797*** (0.409)	-1.931*** (0.408)
Housing: other	-0.572 (0.455)	-0.550 (0.456)	-0.573 (0.454)
Year (2024)	0.294 (0.283)	0.257 (0.286)	0.322 (0.284)
Older adults (age 65+)			-6.043*** (1.696)
Level 2: District characteristics			
Connectivity		-1.068 (0.714)	-1.632* (0.774)
Available for Use		0.210 (0.628)	-0.077 (0.682)
Affordability		-1.343† (0.734)	-1.805* (0.791)
Interactions			
Connectivity x older adults			1.837 (1.817)
Available for Use x older adults			1.314 (1.543)
Affordability x older adults			3.228† (1.851)
Connectivity x lower educational attainment			4.178 (3.370)
Older adults x single-person household			-3.952*** (0.936)
Age x lower educational attainment			-0.277*** (0.080)

Note. Unstandardized coefficients with standard errors in parentheses. Model 0 (null model) had between-district variance = 1.378, within-district variance = 282.100, ICC = 0.49%, and design effect = 2.94. Reference categories: male, college education, white-collar job, apartment, multi-person household, and 2023. †p < .10, *p < .05, **p < .01, ***p < .001.

3.3 Platform Text Analysis: LLM-Assisted Exploratory Case Analysis

Sections 3.1 and 3.2 quantitatively measured Seoul's digital divide using administrative data and survey data and separated the roles of individual characteristics and district-level contextual conditions. However, quantitative indicators alone make it difficult to capture when, and in what contexts, residents experience a lack of digital connectivity in everyday life. This section therefore uses platform text for lived-experience signal detection, meaning the identification of experiences that become visible through users' voluntary posts. The purpose is to contextualize and illustrate the quantitative findings, not to provide confirmatory evidence or population-level prevalence estimates.

The purpose of this section is to use unstructured text from a local community platform to systematically identify platform-visible signals of digital deprivation across the six dimensions of UMC (Connectivity, Available for Use, Affordability, Devices, Digital Skills, Safety) and to present a cautious methodological framework for connecting these signals with the quantitative analyses in Sections 3.1 and 3.2. Rather than treating the platform itself as the object of analysis or as a representative sample of Seoul residents, the section uses platform-visible lived-experience data to supplement the connection between structure and experience.

3.3.1 Characteristics of the Danggeun Platform

The source of the unstructured text is neighborhood-life posts from Danggeun, a Korean local community platform. Danggeun was selected for three reasons. First, each post is automatically assigned geographic metadata at the administrative-dong and district levels, allowing linkage with the UMC index constructed in Section 3.1. Second, platform users are distributed across all of Seoul's 25 districts, which makes district-level aggregation technically possible. Third, because posts voluntarily record problems encountered in everyday life, they can make visible lived-experience signals that surveys may miss. These advantages do not make Danggeun posts representative of Seoul residents as a whole; the analysis observes only the experiences of platform users who chose to post.

3.3.2 Agent-Based LLM Interpretation of Posts

The role of LLMs in this study is not to let a model freely decide what a post means. The research problem is narrower: many posts are written in colloquial Korean, mix slang with local references, and describe everyday situations in which digital connectivity is only one part of the story. A simple keyword search can find candidate posts, but it cannot reliably decide whether digital connectivity is the main problem. Likewise, a topic model can discover themes, but it does not necessarily follow the six UMC dimensions defined in this study. LLMs are used because they can read short everyday texts in context, but their use is made researchable only by fixing what they are allowed to read, what questions they must answer, and how their answers are checked.

Why the interpretation can be treated as a research procedure

A post is not accepted as evidence simply because an LLM produced a label. Each step is tied to an explicit rule. The preprocessing stage first narrows the material with a human-made seed list and a keyword dictionary that is expanded until new keywords become rare. The classification stage then asks the same relevance questions for every post: whether the main issue is digital, whether the post would still make sense if digital words were removed, and whether solving the problem would require a change in infrastructure, devices, skills, affordability, availability, or security. The interpretation stage adds a further requirement: every claim must be linked to a short piece of textual evidence, and alternative explanations such as convenience, ordinary social interaction, or general service dissatisfaction must be considered before a post is treated as a UMC case.

This design is similar to a structured qualitative coding procedure. The prompts function as a codebook, not as open conversation. They define the categories, require the same checks in the same order, prohibit unsupported inferences, and produce an audit trail. Because the output is stored in a fixed format, a human reviewer can see why a post was retained, what dimension was assigned, what evidence was used, and whether the judgment depended on the post itself or on later contextual information.

few-shot classification and reasoning-guided interpretation

The study uses two different prompt designs because it asks two different kinds of questions. Few-shot prompting is used for classification. At that stage, the task is to sort many posts into predefined decisions such as Y, N, or ? and then into the six UMC dimensions. Short examples are useful here because they show the boundary between similar cases:

for example, a post about a high telecom fee is an affordability case, but a post about an expensive restaurant is not; a post about a phishing link is a safety case, but a secondhand trade dispute is not necessarily a cybersecurity case. The examples teach the model the practical boundary of the category. Reasoning-guided prompting is used later because the question has changed. Once a post is already judged to be relevant, the study asks what kind of absence or pathway failure the post suggests. That cannot be answered by examples alone, because two posts in the same UMC dimension may imply different mechanisms. The reasoning prompts therefore require the model to explain the path from the post to the interpretation: what is surprising, what condition would make the situation intelligible, what would normally have happened, where the user's pathway broke down, and what competing explanation might be more plausible.

multiple agents

The three inference agents are best understood as three reading lenses applied to the same post under the same restricted information set. The abductive lens starts from an observed situation and asks what underlying condition could explain it. The forward lens starts from a possible condition and asks what signs should appear in the text if that condition is present. The sequential lens reconstructs the user's pathway and asks where the pathway failed. These are not three votes from identical models. They are three differently constrained readings of the same textual evidence, and their independence matters because the final judgment is stronger when different routes point to the same interpretation and more cautious when they disagree.

Context is managed through a staged input-output structure. First, each hypothesis agent receives only the post text, basic metadata, and the general coding scheme needed to produce a hypothesis; district profiles, living-population context, the absence typology, active category definitions, and the other agents' outputs are deliberately withheld. At synthesis, the judgment synthesizer receives the three hypothesis outputs together with the district context, the absence typology, and the active category set, and then evaluates convergence, disagreement, non-structural alternatives, and the final code. In probability-space terms, this can be read as a staged conditioning design rather than a claim that each post has a fully estimated numerical posterior. The hypothesis agents generate outputs conditional on a restricted information set (I_R), while the judgment synthesizer evaluates the classification conditional on an expanded information set (I_J). This separation reduces leakage from contextual priors into the first reading of the text and makes the later use of context explicit and auditable. Figure 7 visualizes this staged information-set structure.

The full Section 3.3 pipeline, from raw platform records and deterministic filtering to LLM-assisted relevance coding, restricted agent interpretation, judgment synthesis, and district-dimension aggregation, is summarized in Figure 6.

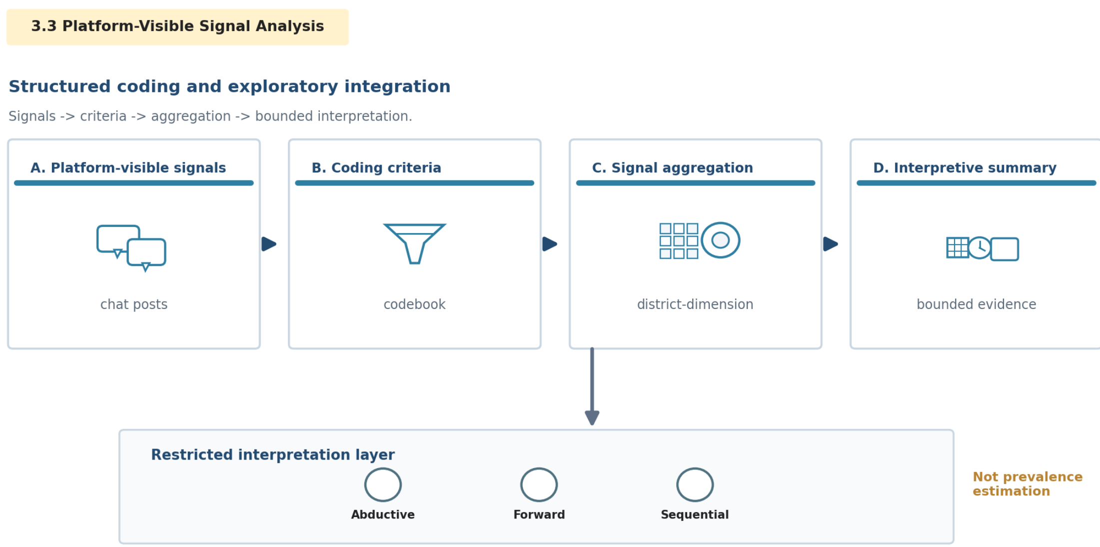


Figure 6. Section 3.3 Platform-Visible Signal Analysis and Bayesian Aggregation Pipeline

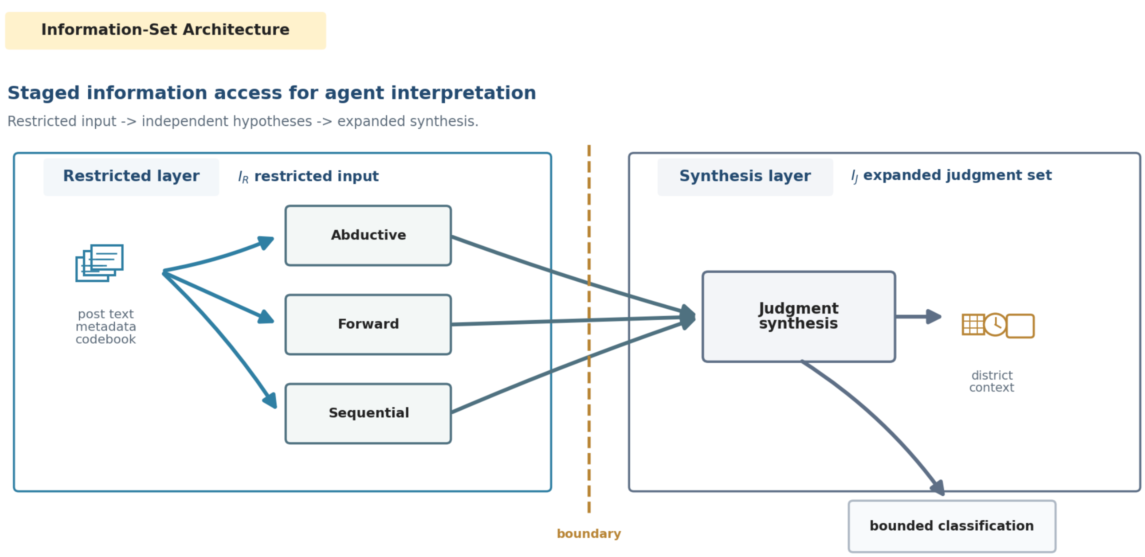


Figure 7. Staged Information Sets for Agent-Based Interpretation

3.3.3 Analytic Structure

The downstream interpretation segment is organized as two linked layers rather than a single gate sequence. Filtered candidate posts first enter UMC relevance and dimension classification. Related posts are then summarized at the district-dimension level through exploratory Bayesian updating, while the agent-based interpretation layer reads post-level cases through three restricted hypothesis agents and a judgment synthesizer. Figure 8 summarizes this segment. Classification decides whether a post belongs in the candidate signal base; Bayesian aggregation describes aggregate platform-prior divergence; structured inference explains what kind of deprivation or pathway failure a selected post may indicate.

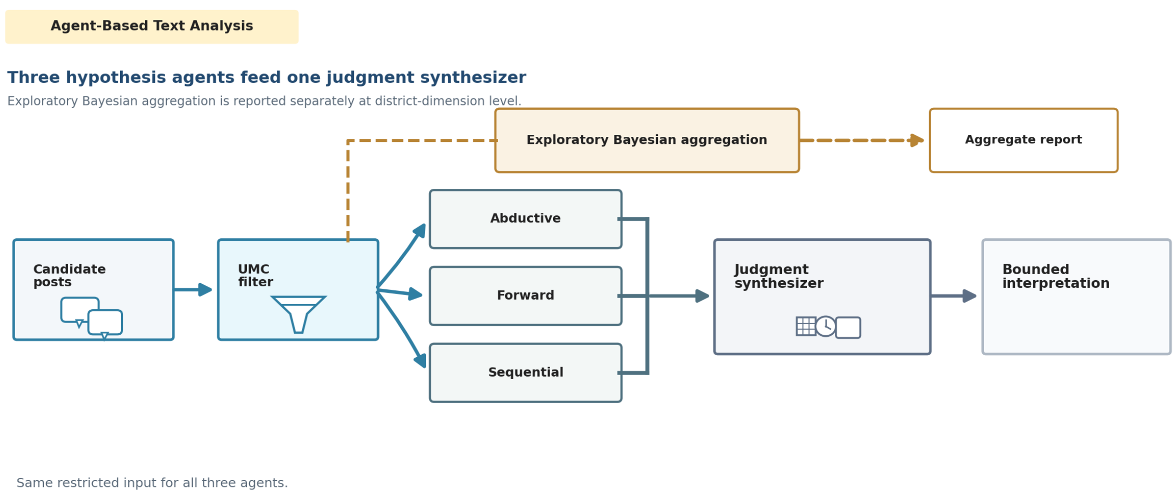


Figure 8. Agent-Based Text Analysis with Parallel Exploratory Bayesian Aggregation

Starting from 1,287,761 raw records, inaccessible body text, duplicates, and advertising posts were removed before keyword screening. After these exclusions, 480,972 posts remained. The keyword dictionary extracted 146,933 keyword-matched candidate posts. Of these, 131,792 posts were successfully classified in the first-stage LLM judgment: 7,136 were classified as related (Y), 2,147 as indeterminate (?), and 122,509 as unrelated (N). Dimension assignment was conducted only for posts classified as related. Bayesian aggregation summarizes their district-dimension distribution and is reported alongside inference outputs rather than used as a filter on the inference agents. Because a single post could be assigned to more than one UMC dimension, these related posts produced 7,421 basic district-dimension signal assignments in the unweighted dimension summary. The later administrative-dong sensitivity table records 7,454 all-weighted mapped assignments after candidate expansion and weighting for ambiguous dong mappings; the two counts therefore refer to different aggregation stages.

To make the first-stage LLM coding auditable, all posts classified as related (Y, 7,136 rows) and unrelated (N, 122,509 rows) were separated into exhaustive internal human-review files. The review layer preserves the original LLM relevance label, dimension assignment, and problem summary while adding separate human-decision columns. The reviewer files contain only the row-level information needed for internal checking and are not reproduced in the report. Public reporting is limited to aggregate validation counts and, after review is completed, to the stability of Bayesian district-dimension results after human-audited labels are incorporated.

The distinction between few-shot classification and reasoning-guided inference is therefore procedural as well as conceptual. Few-shot classification uses examples to keep the boundary of the category stable across many posts. Reasoning-guided inference asks for an explanation of why the selected post should be read as a digital deprivation case. The former is a sorting task; the latter is an interpretation task.

Table 5. Text Preprocessing and Sample Construction

Stage	Description	Count / Output
Raw Data	All Neighborhood Life posts written between May 2024 and March 2026	1,287,761
Accessible Posts	Removed posts whose body text could not be checked because of account withdrawal or blocking	624,564
Duplicate and Advertisement Removal	Removed duplicate content and advertising or promotional posts	480,972
UMC Keyword Filter	Extracted keyword-matched candidate posts by applying dictionaries for the six UMC dimensions	146,933 candidates
First-Stage Judgment	Applied few-shot relevance classification to keyword-matched candidates	131,792 classified; 7,136 Y, 2,147 ?, 122,509 N
Dimension Assignment	Assigned UMC dimensions to posts classified as related; multi-label assignment was allowed	7,421 district-dimension signals

After classification, the district-level distribution of post dimensions is aggregated and compared with the UMC index results from Section 3.1. Bayesian updating is used to combine quantitative prior information with platform-visible signals. The prior represents the district-dimension condition measured by the UMC index, while the new text evidence is treated as a signal that may shift interpretation when enough posts point in the same direction. The report-ready table uses a fixed prior strength, and mapping sensitivity checks compare all-weighted, high-confidence-only, and ambiguous-excluded specifications so that small or ambiguous spatial assignments are not over-interpreted.

$$\alpha_{jd} = \kappa \times s_{jd}$$
$$\beta_{jd} = \kappa \times \left(1 - s_{jd}\right)$$
$$\theta_{jd} | y_{jd} \sim Beta\left(\alpha_{jd} + y_{jd}, \beta_{jd} + n_{jd} - y_{jd}\right)$$
$$n^*_{jd} = n_{jd} \times \left(livingpopdistrict_j / mean\_livingpop\right)$$

The inference component is reported as a post-level interpretive sample rather than as a Bayesian gate output. The abductive, forward, and sequential agents form separate hypotheses under a restricted information set, and the judgment synthesizer compares their outputs with additional context before assigning a final interpretation. The unit of observation is an individual post, while the unit of policy interpretation in the Bayesian layer is the district-dimension combination. Table 6 clarifies these units.

Table 6. Analysis Units and Interpretation Units in Section 3.3

Category	Unit	What It Shows	Interpretive Cautions
Observation Unit	Individual Post	Clues to experiences of digital connectivity	Regional structure cannot be asserted on the basis of a single post alone
Pattern Unit	Administrative-dong Posts	Expression ratios and distributions by dimension	Aggregation criteria and sample bias must be reviewed together
Structural Unit	UMC Regional Indicators	Regional-level connectivity conditions	Interpret only in a limited manner through consistency with the aggregated text results
Causal-Mechanism Unit	Inference Mechanism	Explains the connection between cases and regional patterns	Interpret only within the bounds permitted by the quantitative results in 3.2

3.3.4 Data and Scope of Application

The raw data consist of a crawl of all Neighborhood Life posts written for Seoul's 25 districts from May 2024 to March 2026, totaling 1,287,761 posts after duplicate removal. Of these, 766,891 posts (59.6%), comprising 663,163 deleted posts (51.5%) and 103,728 blocked posts (8.1%), were excluded from the analysis because their body text could not be checked. The final analytic sample consists of approximately 520,000 posts.

A review of the possibility of systematic bias from sample attrition showed that the standard deviation of removal rates across districts was 4.6%, indicating that variation across districts was not large. Categories that were central to inferring digital deprivation, such as Life/Convenience (47.0%), Neighborhood Incidents and Accidents (48.4%), and Hospitals/Pharmacies (35.5%), had lower removal rates than categories oriented toward socializing or promotion, such as Neighborhood Friends (86.6%) and Education (69.0%). A substantial share of deletions appears to have resulted from batch deletions associated with user withdrawal. Nevertheless, it cannot be completely ruled out that some deleted posts included cases related to digital deprivation, or that the act of deletion itself was systematically associated with digital deprivation. This constitutes a limitation of the present analysis.

The structural limitations of the Danggeun data must also be recognized. Because platform users themselves are limited to groups with digital access, the experiences of digitally excluded groups are structurally underrepresented. Since the temporal and spatial distribution of posts is not uniform, sample bias may affect comparisons across districts. These limitations mean that the text-based results should be interpreted as platform-visible lived-experience signals rather than population-level prevalence estimates. In contexts where more direct qualitative data are available, the same pipeline could be applied to produce a more precise analysis.

For the region-dimension results reported below, dong-level linkage was handled as a preprocessing step before district aggregation. All 131,792 classified rows were matched to administrative-dong candidates, and ambiguous legal-dong mappings were retained with equal allocation weights rather than forced into a single dong. This preserves aggregate post mass without turning mapping precision into a substantive finding.

The living-population linkage was checked at the exact date-hour-administrative-dong level to keep the aggregation basis consistent. The resulting density input contains 145,826 exact-matched rows across 420 administrative dongs and 580 dates. Two official Seoul living-population cells were absent from the September 2025 source file and were excluded without interpolation. These checks support the all-weighted Bayesian summary used below; they are quality controls rather than the core result.

3.3.5 Analysis Results

KEY FINDING
Bayesian updating identifies the largest platform-prior divergences in Devices and Digital Skills. Jungnang-gu records the largest positive posterior shifts in Devices (+4.77 percentage points) and Digital Skills (+4.69 percentage points), while Gwanak-gu Digital Skills (-4.52) and Yongsan-gu Devices (-4.18) show the largest negative deviations. These are exploratory platform-visible deviations from the measured UMC prior, not prevalence estimates or proof that the administrative indicators are wrong.

Figure 9 reports the substantive Bayesian result: where platform-visible signals diverge most strongly from the measured UMC prior by district and dimension. Panel A shows the signed posterior shift for all 150 district-dimension cells; Panel B and Panel C isolate the largest positive and negative deviations; and Panel D summarizes which dimensions have the largest average absolute deviations.

At the dimension level, Devices and Digital Skills drive the pattern. Devices has the largest mean absolute posterior shift (2.12 percentage points), followed by Digital Skills (1.55 percentage points). Safety & Security is a secondary source of divergence, while Connectivity, Available for Use, and Affordability show smaller average shifts.

The largest positive deviations are concentrated in Jungnang-gu, especially Devices (+4.77 percentage points), Digital Skills (+4.69), and Safety & Security (+1.78). Songpa-gu also shows a large positive Devices shift (+2.55), and Dobong-gu shows a positive Safety & Security shift (+1.47). These cases indicate where platform-visible help-seeking or inconvenience signals are stronger than the measured prior would imply.

The largest negative deviations appear in Gwanak-gu Digital Skills (-4.52 percentage points), Yongsan-gu Devices (-4.18), Yeongdeungpo-gu Digital Skills (-3.62), Jung-gu Digital Skills (-3.46), Dongjak-gu Devices (-3.31), and Mapo-gu Digital Skills (-3.15). These cases should not be read as absence of need; they indicate that related platform-visible signals are weaker than the measured prior in those district-dimension cells.

Aggregated by district, Jungnang-gu is the clearest upward case, with positive shifts across all six UMC dimensions and an average shift of +2.02 percentage points. By contrast, Mapo-gu, Yeongdeungpo-gu, and Gwanak-gu show broad negative average shifts. This district-level summary helps identify where additional qualitative review is most useful, while keeping the interpretation at the level of platform-visible signals.

Figure 10 maps the all-weighted posterior shifts to show how these region-dimension deviations are spatially distributed. The map does not change the evidence status of the estimates; it helps locate where the largest platform-prior divergences appear across Seoul.

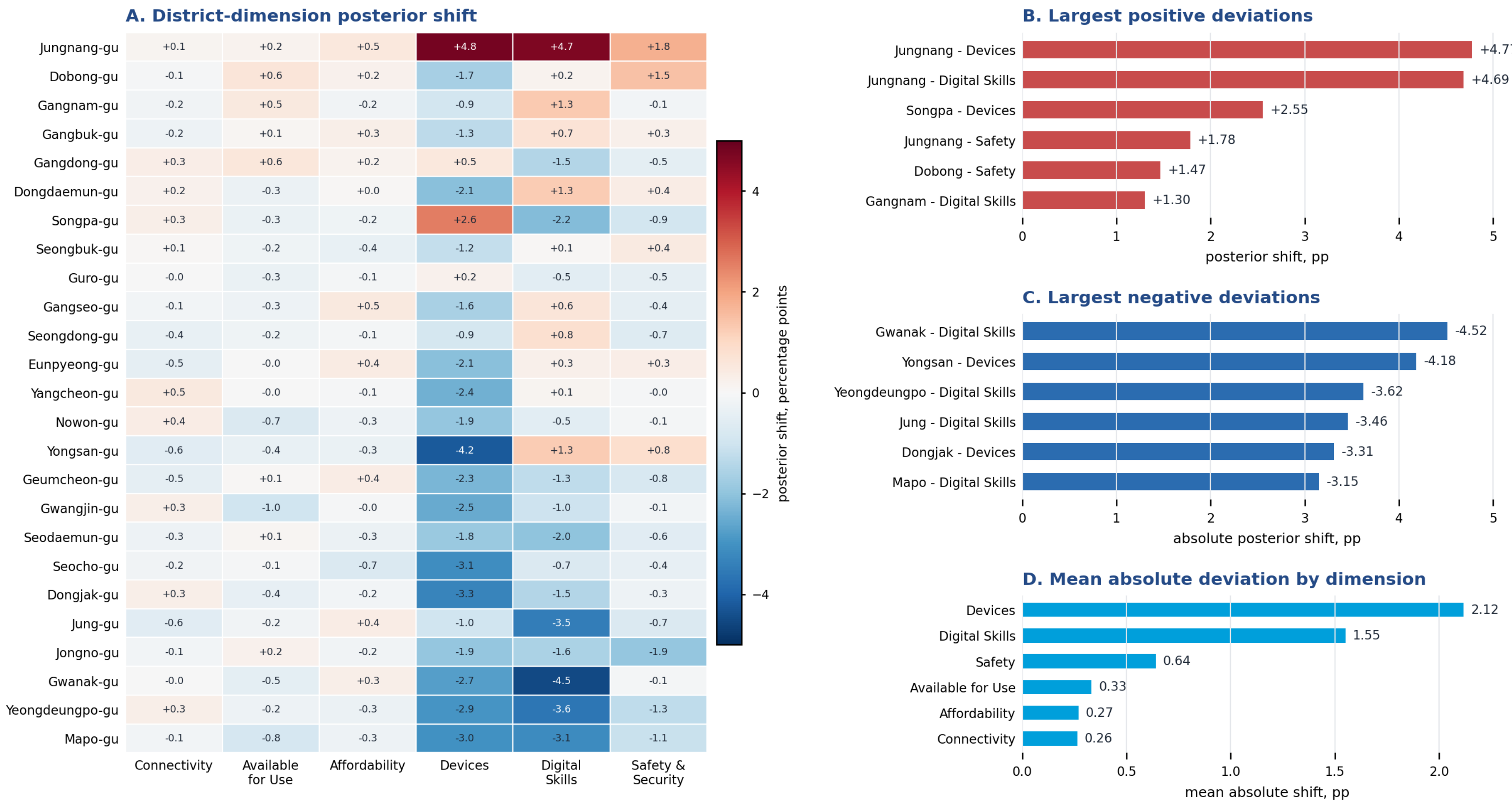
The administrative-dong matching and living-population linkage are treated as preprocessing quality controls. All 131,792 classified rows were matched to administrative-dong candidates, and 149 of 150 district-dimension cells retained the same shift direction across mapping specifications; this supports the stability of the reported pattern but is not the substantive finding.

The categories in Table 8 can be read through several anonymized, platform-visible cases that were already discussed in the analysis. Information absence appears in a case where a resident could not identify where official documents could be printed, even though relevant public-document and self-service issuance channels may have existed. Lack of institutional pathways appears when users leave formal service channels and ask the neighborhood community how to solve a digital task, such as finding the correct customer-service route or the proper redress channel after a digital transaction problem. Lack of device access and lack of public access points appear in posts where a user could not complete a document task because a personal device was unavailable or malfunctioning and therefore searched for a nearby place to use a scanner, printer, or internet-connected computer. Safety and security deficiency appears in platform-visible accounts of secondhand electronic-device transaction harm, phishing, or voice-phishing risk, where the problem is not only awareness of danger but also the capacity to verify information and find a formal response route. These examples show why the high frequency of overlapping categories should be interpreted as a recurring pathway in the exploratory coding subset: service invisibility, weak institutional navigation, insufficient device or access conditions, and safety uncertainty often appear together. They do not estimate population prevalence, but they clarify why the policy interpretation later emphasizes discoverability, institutional navigation, device support, public access-point guidance, and fraud-prevention capability.

Table 7. Post Classification Results

District	Total Posts (n)	Y Classification	? Classification	N Classification	Y Rate (%)
Jungnang-gu	5,476	794	28	4,654	14.5
Songpa-gu	9,367	860	99	8,408	9.18
Seocho-gu	6,170	434	216	5,520	7.03
Eunpyeong-gu	6,140	429	50	5,661	6.99
Dobong-gu	3,491	233	5	3,253	6.67
Seodaemun-gu	4,500	289	5	4,206	6.42
Gangbuk-gu	3,885	247	31	3,607	6.36
Yongsan-gu	3,012	181	12	2,819	6.01
Yangcheon-gu	4,975	296	7	4,672	5.95
Gangdong-gu	6,254	342	231	5,681	5.47
Yeongdeungpo-gu	5,964	306	5	5,653	5.13
Dongdaemun-gu	4,754	238	29	4,487	5.01
Seongbuk-gu	5,437	266	141	5,030	4.89
Jung-gu	1,351	63	16	1,272	4.66
Gangseo-gu	8,174	371	236	7,567	4.54
Geumcheon-gu	2,298	101	7	2,190	4.4
Jongno-gu	2,267	97	27	2,143	4.28
Dongjak-gu	3,675	150	22	3,503	4.08
Gangnam-gu	10,047	404	459	9,184	4.02
Seongdong-gu	404	16	9	379	3.96
Gwanak-gu	11,665	407	75	11,183	3.49
Mapo-gu	7,974	242	254	7,478	3.03
Guro-gu	4,082	121	96	3,865	2.96
Gwangjin-gu	5,440	147	42	5,251	2.7
Nowon-gu	5,040	103	46	4,891	2.04

Where platform-visible signals diverge from measured UMC priors



District	Connectivity	Available for Use	Affordability	Devices	Digital Skills	Safety & Security
Jungnang-gu	+0.1	+0.2	+0.5	+4.8	+4.7	+1.8
Dobong-gu	-0.1	+0.6	+0.2	-1.7	+0.2	+1.5
Gangnam-gu	-0.2	+0.5	-0.2	-0.9	+1.3	-0.1
Gangbuk-gu	-0.2	+0.1	+0.3	-1.3	+0.7	+0.3
Gangdong-gu	+0.3	+0.6	+0.2	+0.5	-1.5	-0.5
Dongdaemun-gu	+0.2	-0.3	+0.0	-2.1	+1.3	+0.4
Songpa-gu	+0.3	-0.3	-0.2	+2.6	-2.2	-0.9
Seongbuk-gu	+0.1	-0.2	-0.4	-1.2	+0.1	+0.4
Guro-gu	-0.0	-0.3	-0.1	+0.2	-0.5	-0.5
Gangseo-gu	-0.1	-0.3	+0.5	-1.6	+0.6	-0.4
Seongdong-gu	-0.4	-0.2	-0.1	-0.9	+0.8	-0.7
Eunpyeong-gu	-0.5	-0.0	+0.4	-2.1	+0.3	+0.3
Yangcheon-gu	+0.5	-0.0	-0.1	-2.4	+0.1	-0.0
Nowon-gu	+0.4	-0.7	-0.3	-1.9	-0.5	-0.1
Yongsan-gu	-0.6	-0.4	-0.3	-4.2	+1.3	+0.8
Geumcheon-gu	-0.5	+0.1	+0.4	-2.3	-1.3	-0.8
Gwangjin-gu	+0.3	-1.0	-0.0	-2.5	-1.0	-0.1
Seodaemun-gu	-0.3	+0.1	-0.3	-1.8	-2.0	-0.6
Seocho-gu	-0.2	-0.1	-0.7	-3.1	-0.7	-0.4
Dongjak-gu	+0.3	-0.4	-0.2	-3.3	-1.5	-0.3
Jung-gu	-0.6	-0.2	+0.4	-1.0	-3.5	-0.7
Jongno-gu	-0.1	+0.2	-0.2	-1.9	-1.6	-1.9
Gwanak-gu	-0.0	-0.5	+0.3	-2.7	-4.5	-0.1
Yeongdeungpo-gu	+0.3	-0.2	-0.3	-2.9	-3.6	-1.3
Mapo-gu	-0.1	-0.8	-0.3	-3.0	-3.1	-1.1

Figure 9. Largest Posterior Shifts Between UMC Prior and Platform-Visible Signals by District and Dimension

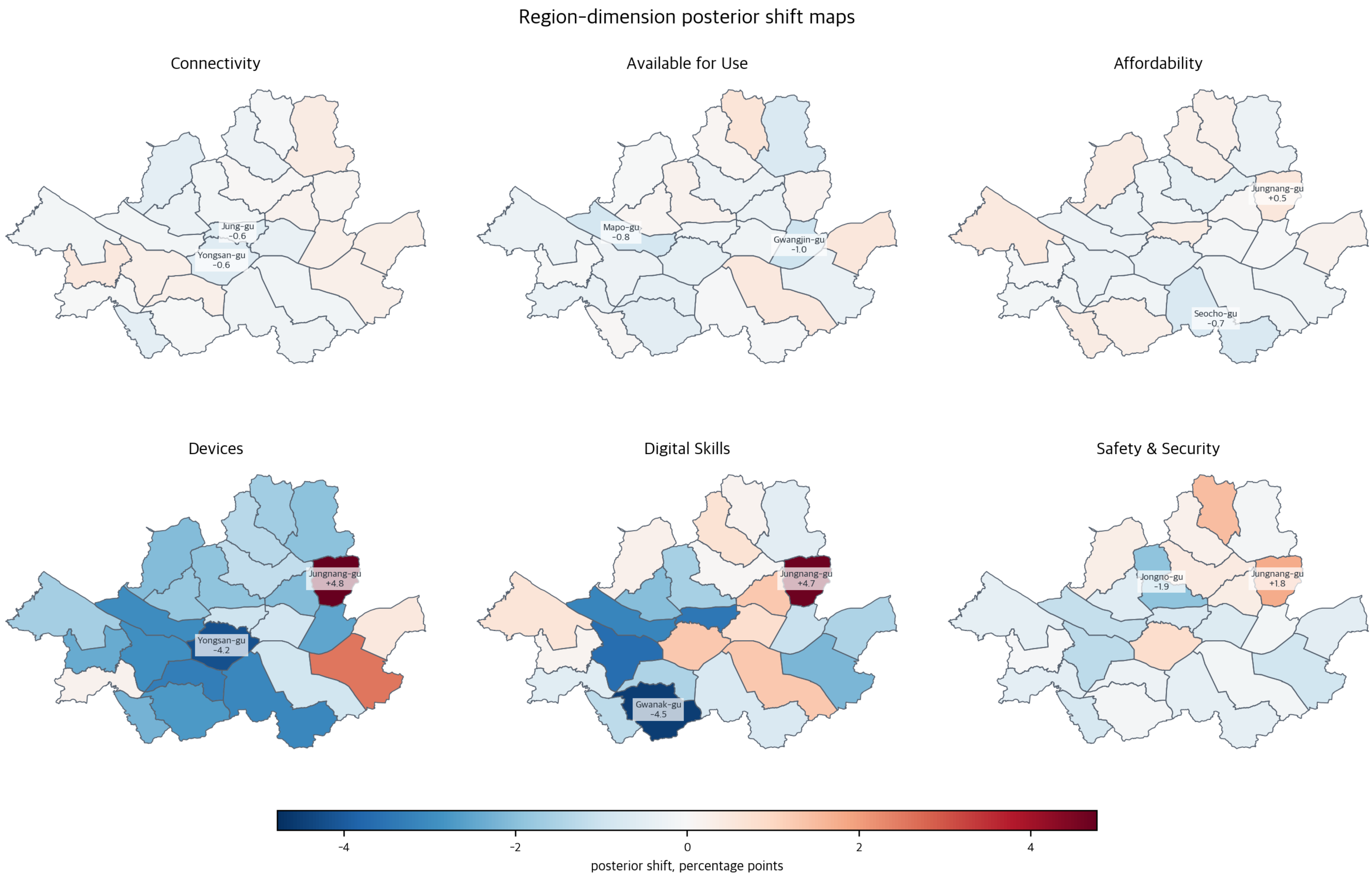


Figure 10. Spatial Distribution of Region-Dimension Posterior Shifts

Table 8. Hypothesis-generation results: distribution by type of deprivation

Type of deprivation	Related UMC dimension	Number of related posts (n=100)	Share (%)
Information absence	Digital Skills	63	63.0
Lack of institutional pathways	Available for Use	61	61.0
Lack of device access	Devices	50	50.0
Absence of digital competency	Digital Skills	45	45.0
Lack of connectivity infrastructure	Connectivity	40	40.0
Lack of public access points	Available for Use	32	32.0
Lack of economic access	Affordability	27	27.0
Safety and security deficiency	Safety	16	16.0
Composite deprivation (two or more)	—	88	88.0

Note. Because multiple types of deprivation may be inferred from a single post, the sum of the percentages exceeds 100%. The first agent layer is a hypothesis-generation stage, and final classification requires triangulation and human validation before the results can be used as confirmatory evidence.

3.4 Synthesis of Analytical Results

KEY FINDING
The three evidence layers point to the same policy logic. The index identifies where district conditions differ, the HLM shows that individual vulnerabilities explain most digital-use inequality, and platform-visible/Bayesian signals identify district-dimension mismatches and lived-experience pathways. No layer alone is treated as confirmatory evidence.

This section synthesizes the results of Section 3.1 (construction of the UMC index), Section 3.2 (HLM multilevel analysis), and Section 3.3 (LLM-assisted platform-visible signal analysis) to provide an integrated interpretation of Seoul's digital divide. The three analyses respectively answer the questions of 'where relative district conditions differ' (measurement), 'which individual and contextual factors are associated with digital usage' (explanatory decomposition), and 'how digital disadvantages become visible in everyday situations' (exploratory qualitative context). Policy-relevant implications emerge at their points of intersection, but each evidence layer retains its own limits.

The first and central empirical result of this project is that Seoul's digital divide is structured more strongly by differences between individuals than by differences across districts. In the HLM analysis in Section 3.2, the ICC was only 0.49%, meaning that 99.5% of the variance in digital usage scores was attributable to differences between individuals within the same district. This result indicates that, although the cross-district variation in the UMC composite index constructed in Section 3.1 is real, its association with individual digital usage is limited as a broad average pattern. This low ICC does not make district-level variables irrelevant; rather, it means that they should be interpreted as contextual conditions and targeting aids, especially when combined with individual vulnerability profiles.

The text analysis in Section 3.3 contextualizes how this quantitative finding becomes visible in everyday life. The fact that 'information absence' appeared frequently in the exploratory platform sample suggests that, among some users, the existence of digital services may not be recognized even when services are formally available. Cases in which users did not know about self-service printing at convenience stores, did not know the ISP customer center number, or did not know about Government24's public-document printing function all point to situations in which 'services exist but are not visible to users.' This should be read as illustrative and contextual evidence that helps explain why the education gradient in the HLM may matter in practice, not as proof of the multilevel model.

The second finding is that the associations of district-level infrastructure are not robust as broad average patterns. Model 2 shows no statistically significant positive main effect for Connectivity or Available for Use, and Model 3 does not provide strong evidence that these infrastructure dimensions significantly moderate digital usage for older adults or lower-education residents. The results therefore support a cautious interpretation: infrastructure conditions matter as part of the local policy environment, but they should not be treated as a standalone explanation for Seoul's digital divide.

The text analysis in Section 3.3 provides qualitative context for this limited infrastructure finding. The exploratory inference that public access-point issues appeared in a subset of posts suggests that, even when infrastructure physically exists, it is not converted into substantive benefits unless users' awareness and access capabilities are also present. The case from Sinsa-dong, Eunpyeong-gu ('Could it really be that there is none?') illustrates the possibility that the user had already attempted an online search but found the results insufficient, thereby highlighting the gap between the existence of infrastructure and its discoverability.

The third finding is that the digital divide is deepened by intersections of complex vulnerability rather than by a single axis. In the HLM, older adults living alone scored an additional 4.0 points lower than older adults not living alone, and the negative association between lower educational attainment and digital use was found to accelerate as age increased. Section 3.3 likewise illustrates a pattern of compound deprivation in platform-visible cases. When information absence, competency absence, and blocked institutional pathways overlapped, users were observed to leave official channels and rely on the community. The absence of a digital support network ('because they do not have children, or because they live apart from them and find it difficult to ask') illustrates how household composition can function as an informal pathway for acquiring digital competency.

The fourth finding concerns the spatial clustering of provisionally defined Digital Desert districts and heterogeneity by dimension. In Section 3.1, among the bottom five areas, northern Seoul in particular (Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu) formed a Low-Low cluster in the UMC composite index; while these areas recorded above the city average on the Connectivity dimension, they showed low scores in the Devices and Safety dimensions. This heterogeneity in dimension-specific profiles suggests the need for dimension-specific, place-sensitive support rather than uniform infrastructure investment or fixed labels.

From a methodological perspective, combining the three analyses allows their respective limitations to be made explicit rather than hidden. The composite index in Section 3.1 identifies the relative positions of districts but cannot capture within-individual variation, while the HLM in Section 3.2 separates the contributions of individuals and areas but cannot explain the qualitative pathways through which the digital divide is manifested. The text analysis in Section 3.3 makes visible the experiential context of deprivation but is limited in representativeness and validation. By combining the three analyses through a sequential structure rather than in parallel (measurement -> explanatory decomposition -> exploratory qualitative context), the project constructs an integrated answer to the three questions of 'where' (districts), 'for whom' (individual vulnerability), and 'how' (the pathways through which deprivation is made visible).

These synthesized results lead to the policy recommendations in Chapter 4. The finding that individual characteristics are the dominant explanatory variables provides the basis for prioritizing individual-vulnerability-centered support; the limited and context-dependent role of district UMC dimensions supports place-sensitive targeting rather than infrastructure-only policies; the observed pattern of complex vulnerability indicates the need for an integrated support system rather than a single program; and the identification of spatial clustering provides a practical basis for inter-district coordination without treating districts themselves as causes.

The HLM results identify the dominant structure of digital usage inequality, while the platform-visible signal analysis contextualizes how such inequality becomes visible in everyday situations. The text-based evidence does not prove the multilevel model, nor does it estimate population prevalence. Instead, it supplements the quantitative findings by revealing concrete pathways through which digital disadvantages are experienced, narrated, and made visible on a local platform.

4. Policy Recommendations

This chapter presents policy directions for addressing Seoul's digital divide based on the analytical results in Chapter 3. The data used in the analysis were collected in 2023-2024 and therefore reflect the policy environment under the Framework Act on Intelligent Informatization. As of January 2026, the Digital Inclusion Act came into effect, shifting the policy frame from the previous approach centered on closing the information gap to the expanded concept of digital inclusion. The recommendations in this chapter present the gaps and vulnerabilities identified under the previous system as observational evidence, while exploring their potential policy application within the new legal framework.

4.1 Theoretical Framework for Policy Recommendations

Policy approaches to Seoul's digital divide should be organized around individual vulnerability first and place-sensitive targeting second. In Section 3.2, Model 1, which included only individual-level variables, explained 83.4% of the variance between districts; this means that most of the cross-district gap in digital usage is attributable less to intrinsic local characteristics than to differences in the sociodemographic composition of residents in each district. Accordingly, the primary route of policy intervention should directly support groups with multiple vulnerability factors, especially older adults with lower educational attainment, older adults living alone, groups with limited experience using digital services, and residents with weak offline access to assistance.

However, place-sensitive targeting must be distinguished from uniform resource allocation at the district level. The ICC of 0.49% indicates that variance between districts is not the main source of the digital usage gap, suggesting that policies centered only on area-level allocation may be inefficient. What this project proposes is a place-sensitive approach: a strategy that uses the UMC composite index and dimension scores to identify where vulnerable individuals may be reached more efficiently, while improving spatial conditions that make services easier to find, reach, and use. The HLM results do not justify treating public Wi-Fi density or connectivity as a clear standalone multiplier, but they do show that district-level contextual conditions should be interpreted in combination with individual vulnerability and platform-visible evidence on discoverability.

This dual approach is also consistent with internationally accepted policy frameworks. OECD (2023) emphasizes that, under a structure of widening interregional gaps (uneven divergence), place-based policies and people-centered transfer policies are not mutually exclusive but complementary, while ITU (2024)'s UMC policy agenda likewise recommends both target-group-centered policies and the development of national digital strategies and roadmaps. EU Digital Decade 2030 establishes national roadmaps for each member state across four axes - digital skills, infrastructure, business, and public services - and operates a cyclical structure in which progress is reviewed through annual monitoring (European Commission, 2025). This structure of measurement -> monitoring -> policy adjustment partly aligns with the analytical procedure of this project.

4.2 Individual-Vulnerability-Centered Support

(1) Older adults living alone

The predicted digital usage gap for older adults living alone is approximately -10.87 points, corresponding to 13.6% of the full scale. Whereas older adults in multi-person households can receive informal digital support from household members, this pathway is absent for older adults living alone. The platform-visible signal analysis in Section 3.3 illustrates how this gap can appear in everyday life. The case of an IT planner in Munjeong-dong, Songpa-gu, who proposed app-use training for middle-aged and older adults after observing that a father could not use mobile-payment services, illustrates that one-time instruction is often insufficient and that repeated, relationship-based support is needed.

Existing Digital Learning Centers have centered on group education and have faced structural limitations in physically reaching older adults living alone whose mobility is constrained. With the transition to AI Digital Learning Centers from March 2026, outreach dispatch education is expected to expand (Ministry of Science and ICT, 2025.12.24), but program designs specialized for older adults living alone have not yet been specified. The United Kingdom's Age UK Digital Champion Programme has reported improvements in older adults' digital confidence by combining volunteer-based one-on-one home visits with device lending (Age UK, 2023).

Taken together, policy for older adults living alone is insufficient if it only expands access to group education; what is needed is a model that substitutes for social scaffolding. Specifically, this report recommends a one-on-one home-visit support system through the cultivation of dong-level Digital Companions, the operation of cohorts tailored to older adults living alone, and the parallel provision of device lending.

(2) Older adults with lower educational attainment

In M3, the age x lower educational attainment interaction (-0.28, $p < 0.001$) shows that the age-specific slope of decline in digital usage for the lower-educational-attainment group is approximately twice as steep as that for the college-graduate reference group. Over a 20-year age range, this difference accumulates into an additional gap of approximately 5.5 points, suggesting that the digital divide between educational groups tends to widen as age increases.

The platform-visible signal analysis in Section 3.3 illustrates how this education gradient may appear in everyday life. The fact that 'information absence' appeared frequently in the exploratory sample suggests that, as educational level decreases, the gap in awareness of the very existence of digital services may intensify. Cases in which users did not know about self-service printing at convenience stores, did not know the ISP customer-service telephone number, or did not know about Government24's public-document printing function all show how services can remain invisible even when they formally exist.

It is unclear whether existing Digital Learning Centers systematically provided differentiated pathways by educational level. Singapore IMDA's Seniors Go Digital operates an integrated model that establishes a three-stage pathway of basic, intermediate, and advanced levels, embeds cybersecurity education at each stage, and combines device and fee subsidies for low-income older adults (IMDA, 2020; CSA Singapore, 2023). Security education is particularly relevant in the Seoul context, where vulnerability in the Safety dimension was identified in Section 3.1.

The recommendations are to systematically differentiate staged pathways by educational level, to adopt an integrated design that makes security education mandatory at each stage, and to link device and fee support for low-income older adults with lower educational attainment.

(3) Strengthening the discoverability of digital services to address information absence

One policy task suggested by the platform-visible signal analysis in Section 3.3 is that existing digital competency education has focused on 'how to use' services, whereas the gap in 'awareness of existence' has not been sufficiently addressed as a policy issue. Information absence, inferred in 63 of the 100 exploratory cases analyzed, refers not to a lack of capability to use services but to a condition in which the existence of the services themselves is unknown.

For example, in the case of a Bongcheon-dong resident in Gwanak-gu who asked, 'It seems that court documents and the like cannot be printed at ordinary print cafes; does anyone know where official documents can be printed?', neither the Government24 official-document printing function nor the existence of court self-service issuance kiosks was visible to the user. The expression 'Is there truly none?' used in Sinsa-dong, Eunpyeong-gu suggests that the user may already have attempted an online search but found the results insufficient, highlighting the gap between the existence of infrastructure and its 'discoverability.'

This absence of information is difficult to resolve simply by strengthening publicity. Passive provision of information reaches only users who already possess the capacity to search for information. Therefore, as a policy design principle, an approach is needed that embeds the discoverability of digital services from the service design stage onward. Specifically, we propose functions in public service apps (such as Seoul ON) that provide location-based guidance to nearby public digital access points (self-service printing, library multifunction printers, and AI Digital Learning Centers), offline linkage that informs residents of relevant digital services when they visit community service centers for civil affairs, and the parallel listing of relevant digital channels in text-message notifications from the competent dong office.

(4) Structural Responses to the Educational Gradient

Educational attainment is the strongest predictor of digital use, and this gap has a structural character that is difficult to resolve through short-term programs alone. It is therefore necessary to distinguish the time horizons of policy. In the short term, the targeted programs described above can contribute to mitigating current gaps; in the medium to long term, systematically incorporating digital competency into curricula from the basic education stage becomes a structural pathway for preventing the cumulative widening of gaps in future generations. The 'matters concerning activities to cultivate digital competency' and 'education, counseling, and publicity for digital inclusion' required in the basic plan under Article 8 of the Digital Inclusion Act provide the institutional basis for encompassing precisely these short-, medium-, and long-term policies at the same time.

4.3 Place-Sensitive Targeting and Dimension-Specific Support

(1) Dimension-Specific Place-Sensitive Support for Provisional Digital Desert Districts

Section 3.1 provisionally classified the five districts at the bottom of the composite index (Jungnang-gu, Dobong-gu, Gangbuk-gu, Guro-gu, and Nowon-gu) as 'Digital Deserts.' This label is used as a diagnostic and targeting device, not as a fixed identity or proof that the districts themselves cause deprivation. Their core characteristic is that, while they record levels at or above the city average in the Connectivity dimension (Jungnang-gu 0.630, Gangbuk-gu 0.562), they exhibit low observed scores in the Devices (Jungnang-gu 0.052) and Safety (Dobong-gu 0.019) dimensions. This suggests that the main policy problem is not only the absence of physical network infrastructure, but also the human, economic, and safety-related capabilities needed to use it.

Because the dimension-specific decomposition (Figure 3) shows that each district has different vulnerable dimensions, dimension-specific support may be more effective than uniform support. For example, device distribution and lending programs should be prioritized in Jungnang-gu, where the Devices dimension is extremely low, while security education and awareness-raising programs should be prioritized in Dobong-gu, where the Safety dimension is effectively close to zero. For districts vulnerable in the Affordability dimension, fee subsidies and expansion of low-cost plans constitute a more direct intervention pathway.

The platform-visible signal analysis in Section 3.3 provides contextual illustrations for dimension-specific support. In the case of Dobong-gu, where the Safety dimension is extremely low, platform-visible posts included cases of harm in online secondhand transactions. The pattern in which the absence of capacity to verify the condition of secondhand electronic devices is combined with a lack of information about formal redress channels after harm occurs (consumer counseling centers, police reports, etc.) suggests that security education should extend beyond simple password management to include consumer protection capacities in digital transaction environments.

Given that the main associations of district UMC dimensions are limited and not uniformly positive, place-sensitive interventions have limits as a standalone means of closing the gap. This means that a more defensible policy design is likely when dimension-specific support is combined with the individual-vulnerability-centered policies discussed in 4.2, for example, when device distribution is accompanied by level-specific education.

It is necessary to examine whether the current policy system allocates resources differentially in ways that reflect the characteristics of each district. The United Kingdom's Digital Inclusion Innovation Fund (GBP 11.7M, 2025) adopted a model that allocates funding to 80 local projects through a community-led approach while differentiating project content according to local needs (older adults, low-income groups, migrants, etc.) (UK DSIT, 2025). This provides a reference case demonstrating the feasibility of differentiated interventions suited to local contexts.

(2) Redesigning Public Digital Access Points

The exploratory inference in Section 3.3 that some posts reflected a lack of public access points raises the possibility that Seoul's public digital infrastructure is not quantitatively insufficient, but rather is designed and located in ways that do not correspond to users' task contexts. In cases such as a student living alone in Sillim-dong, Gwanak-gu asking for a recommendation for 'a PC room where document scanning is possible because mine is not working,' or a Bongcheon-dong resident in Gwanak-gu asking, 'Is there anywhere near Sadang Station where I can use the internet and a printer?', users were not aware of existing public access points such as self-service printing in convenience stores or library multifunction printers.

These findings suggest that policy on public digital access points should shift from 'installation' to 'discoverability and accessibility.' Specifically, we propose establishing an integrated guidance system for public digital access points by district (self-service printing, scanning, public Wi-Fi, and AI Digital Learning Centers), strengthening searchability through data linkages with private platforms such as Naver Map and KakaoMap, and securing access points outside the operating hours of public facilities, including late-night and weekend periods (for example, 24-hour self-service printing booths).

(3) Coalition Governance for Seoul's Northern Districts

The spatial distribution of the UMC composite index (Figure 4) and the LISA cluster analysis (Figure 5) show that provisionally defined Digital Desert districts are clustered in northern Seoul. The physical adjacency of Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu, which belong to the Low-Low cluster, suggests a spatially patterned contextual condition that may justify coordination across neighboring districts.

This spatial clustering provides three grounds for recommending cooperative governance at the inter-district scale. First, the composite-index LISA analysis indicates significant spatial autocorrelation, supporting the interpretation that the northern Low-Low cluster is not merely a visual pattern. Second, if individual districts independently build digital education programs or infrastructure, resources may become dispersed; however, if neighboring districts jointly operate digital hubs or share education programs, economies of scale can be achieved. Third, as shown in Section 3.1, adjacent districts differ in their dimension-specific strengths and weaknesses (e.g., Jungnang-gu's strength in Connectivity at 0.630 versus its weakness in Devices at 0.052), making cross-utilization of resources possible.

Specifically, we propose establishing coalition governance for the northern Digital Desert districts (Dobong-gu, Gangbuk-gu, Nowon-gu, Jungnang-gu, and adjacent partners). This includes operating joint digital education hubs, cross-utilizing dimension-specific strengths and weaknesses, and securing budgets and planning projects at the inter-district scale. The institutional pathway for this cooperation system is discussed in 4.4 in connection with the basic-plan system of the Digital Inclusion Act.

4.4 Integrated Capability, Safety, and Affordability Strategy

The preceding analytical results and policy recommendations can be organized into three linked policy directions. First, individual-vulnerability-centered support should target low-education older adults, older single-person households, residents with low digital service-use experience, and residents with weak offline access to assistance. Second, place-sensitive targeting should use the UMC composite index and dimension scores to identify where these groups may be reached more efficiently, without treating a low district score as proof that the district itself causes deprivation. Third, the emphasis should move beyond infrastructure expansion alone toward capability, safety, and affordability: device access, digital service-use training, security and fraud-prevention capacity, telecom affordability, and help with complex public apps. The platform-visible signal analysis supplements this strategy by illustrating concrete situations in which overlapping barriers become visible, but it does not by itself establish population prevalence or policy effects.

Article 8 of the Digital Inclusion Act stipulates that the government must establish a basic plan for digital inclusion every three years, and that this basic plan must include 'improving accessibility and supporting social participation for digitally vulnerable groups,' 'analyzing the status of digital inclusion and policy performance,' and 'activities to cultivate digital competency,' among other matters. Article 9 of the same Act requires relevant central administrative agencies and local governments to establish and implement annual implementation plans. The results of the multilevel analysis conducted in this project, including the decomposition of between-district and within-district variance, identification of vulnerable groups, and dimension-specific profiling, can be used as observational evidence in the process of formulating this basic plan. In particular, the platform-visible distribution by deprivation type derived from Section 3.3 can inform priority-setting in the basic plan, provided that it is treated as exploratory and contextual rather than confirmatory.

The institutionalization of the coalition governance for the northern Digital Desert districts proposed in 4.3 can also be explored within this system. Although the Digital Inclusion Act designates metropolitan and basic local governments as the actors responsible for establishing and implementing implementation plans, it does not contain separate provisions for cooperation systems among adjacent basic local governments. In the process of formulating the basic plan, it is necessary to review institutional pathways for such cooperation, for example, the creation of consultative bodies for joint project planning among districts.

Finally, from the perspective of procedural generalizability discussed in 1.1, the analytical framework of this project, namely digital divide diagnosis (construction of a composite index and dimension-specific decomposition) -> multilevel analysis (separation of individual and district contributions and identification of vulnerable groups) -> exploratory lived-experience signal detection (LLM-assisted classification of platform-visible deprivation types) -> targeted policy design (integration of individual-vulnerability-centered support and place-sensitive targeting), is significant as a policyderivation procedure that can be applied not only to Seoul-specific prescriptions but also to other cities with similar data structures.

5. Conclusion

5.1 Summary

This project applied the ITU's Universal Meaningful Connectivity (UMC) framework to Seoul's 25 districts and constructed an integrated analytical framework for measuring, analyzing, and interpreting the digital divide within the city. By combining three stages of analysis, construction of the UMC composite index (3.1), HLM multilevel analysis (3.2), and LLM-assisted exploratory text analysis (3.3), the project clarified at multiple levels where Seoul's digital divide is observed, which individual and contextual factors are associated with it, and how it is made visible in platform-mediated everyday situations.

First, the UMC composite index analysis found a gap of up to 2.5 times in digital connectivity across Seoul's districts. The difference between Seocho-gu (0.695) and Jungnang-gu (0.279) shows that, contrary to the general perception of Seoul as a highly connected city, a considerable relative gap is observed at lower-level administrative units. In particular, four of the five provisional Digital Desert districts are spatially clustered in northern Seoul (Dobong-gu, Gangbuk-gu, Nowon-gu, and Jungnang-gu), while Guro-gu appears as a separate southwestern low-score case. These districts exhibit dimension-specific heterogeneity: although the northern cluster is at or above the city average in the Connectivity dimension, it shows low observed scores in the Devices and Safety dimensions.

Second, the HLM analysis indicates that Seoul's digital divide is structured overwhelmingly by individual characteristics rather than by district-level infrastructure differences. The ICC was only 0.49%, indicating that 99.5% of the variance was attributable to differences among individuals, and educational attainment (middle school graduation or less, about -22.6 points in Models 1 and 2) and age (about -0.35 points per year in Models 1 and 2) were the strongest predictors. The current main model does not support the earlier draft's strong claim that public Wi-Fi or connectivity infrastructure produces clearly significant cross-level benefits for older adults or lower-education residents. Instead, the robust interaction evidence is concentrated at the individual level: older adults living alone show an additional gap of about -4.0 points, and the age x lower educational attainment interaction indicates that educational disadvantage widens with age.

Third, the LLM-assisted text analysis used large language models as structured coding assistants to examine 100 Danggeun neighborhood-life posts in the hypothesis-generation stage. Information absence (63%), absence of institutional pathways (61%), and lack of device access (50%) emerged as frequent exploratory categories, and two or more forms of deprivation overlapped in 88% of posts. These figures should not be interpreted as the prevalence of digital deprivation among Seoul residents. Rather, they make visible pathways of deprivation that quantitative analysis cannot capture, such as the stage at which official channels are abandoned and users turn to community support.

Based on these analytical results, Chapter 4 presented policy directions that integrate individual-vulnerability-centered support (older adults living alone, older adults with lower educational attainment, and responses to the educational gradient) and place-sensitive targeting (dimension-specific support for provisional Digital Desert districts and coalition governance for the northern districts). These directions are linked to the basic-plan formulation system under the Digital Inclusion Act, which came into effect in 2026, and the analytical framework itself has procedural generalizability applicable to other cities with similar data structures.

5.2 Limitations and Future Tasks

This project has the following limitations, which indicate directions for future research.

First, there is the limitation of cross-sectional data. The Seoul Survey (2023–2024) used in the HLM analysis and the administrative data used to construct the UMC index capture cross-sections at specific points in time and therefore cannot track temporal changes in the digital divide. Panel data need to be constructed in order to examine how the associations of individual characteristics such as educational attainment or age with digital use change over time, and whether gaps actually narrow after policy interventions. Although the fact that the Seoul Survey is an annually conducted administrative survey suggests the possibility of conversion to a panel structure, the current sample design is a repeated cross-sectional structure and therefore has limits for individual-level tracking.

Second, there are constraints on multilevel association analysis. HLM estimates the direction and magnitude of relationships among variables, but because the analysis is based on observational data, caution is required for strict causal interpretation. For example, the negative district-level coefficients for Connectivity and Affordability in the current main model should not be interpreted as evidence that infrastructure itself reduces digital usage; they may reflect contextual differences, composition effects, measurement limits, or unobserved characteristics of districts. Future research needs to identify causal relationships more rigorously through quasi-experimental designs, such as difference-in-differences, that use policy changes as exogenous shocks.

Two additional HLM interpretation constraints should also be noted. The lower-education subgroup is relatively small, so interaction estimates involving lower educational attainment should be interpreted as identifying a vulnerable pattern rather than as a precise population parameter. The positive coefficient for disability status should also be read cautiously because it may reflect survey selection or accessibility conditions among respondents rather than a general advantage among people with disabilities.

Third, several limitations apply to the platform-visible signal analysis. The platform-visible signal analysis should not be interpreted as estimating the prevalence of digital deprivation among Seoul residents. Rather, it identifies platform-visible lived-experience signals that supplement administrative and survey-based indicators. Danggeun posts do not constitute a representative sample of Seoul residents; they capture only experiences made visible by platform users who choose to post about them. The LLM functions as a structured coding assistant, not as an autonomous source of evidence. Although the prompt-based classification procedure follows a structured prompt that functions similarly to a qualitative codebook, the outputs cannot be treated as equivalent to validated human-coded data. Misclassification may occur in borderline cases, such as ordinary service complaints, sarcastic posts, commercial posts, or cases involving multiple UMC dimensions. The current analysis does not fully report human validation metrics such as precision, recall, F1-score, Cohen's kappa, or Krippendorff's alpha. Precision is the share of LLM-classified relevant cases that are actually relevant; recall is the share of actually relevant cases that the LLM identifies; F1-score combines precision and recall; Cohen's kappa and Krippendorff's alpha evaluate agreement between human and LLM judgments or among human coders. These validation procedures should be conducted in future work before platform text results are used as confirmatory evidence. In this study, the LLM-based analysis is therefore interpreted as an exploratory supplement that contextualizes and illustrates the quantitative findings rather than proving them.

Fourth, there are methodological constraints in constructing the UMC index. This project applied equal weights to the six dimensions, but the relative importance of dimensions may vary by context. For example, in Seoul, where device ownership has become widespread, further review is needed as to whether assigning the same weight to the Devices dimension as in other cities is appropriate. In addition, because Min-Max normalization is sensitive to extreme values, additional verification of index stability is required given the small number of districts (25).

Fifth, spatial resolution remains a methodological constraint, although the present revision extends the platform-text layer to administrative-dong matching. Districts (n=25) are still the main reporting unit for the integrated UMC interpretation, but the dong-level linkage shows that finer-grained platform signals can be organized when safe metadata and exact living-population keys are available. Future work should develop validated dong-level models and compare them with district-level results, especially because two official living-population cells were missing and ambiguous legal-dong mappings require sensitivity checks.

Sixth, the Digital Desert label is provisional and diagnostic. It helps identify districts where multiple UMC dimensions appear weak and where place-sensitive targeting may be useful, but it should not be read as a fixed district identity, a stigmatizing classification, or evidence that the district itself causes individual digital disadvantage.

Finally, the analytical framework of this project, which constructs a composite index based on the UMC framework, separating individual- and district-level contributions through a multilevel model, and extracting exploratory platform-visible context using LLM-assisted coding, has value not only as a basis for Seoul-specific prescriptions, but also as a methodological framework applicable to other cities with similar administrative data structures and digital platform ecosystems. To test the external validity of this framework, a future task is to apply it to districts with different levels of urbanization and digital infrastructure environments.

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Appendix. Text Classification and Agent Prompt Materials

This appendix records the prompt materials used to move from raw community posts to auditable UMC interpretations. It includes the text preprocessing dictionary, the post relevance and dimension-classification prompt, and the four structured interpretation prompts used in the inference stage. Post-level identifiers, original post text, and local raw-data paths are not reproduced. Identifier fields are shown only as <POST_ID>, and batch placeholders are shown as {BATCH_NUM}.

A. Text Preprocessing Keyword Dictionary

The text preprocessing stage uses two keyword resources. The initial seed list is intentionally broad and is used to collect candidate posts. The LLM-expanded dictionary is then built from sampled posts and used to tag candidate posts by matched UMC dimensions. The table reports the size and representative content of the dictionary; the complete dictionary is maintained in the project analysis repository and is not reproduced in full to keep the report readable.

Appendix Table A1. Keyword dictionary used for candidate construction

Dimension	Dictionary size	Representative expressions	Use in pipeline
Connection Quality	196	internet disconnection; weak signal; slow Wi-Fi; telecom outage; high ping; unstable data	Quality and stability posts
Availability for Use	157	public Wi-Fi; service unavailable; app-only service; data exhaustion; unsupported area; online-only application	Use-availability posts
Affordability	129	telecom fee; data charge; budget carrier; plan discount; device cost; subscription burden	Service or device cost posts
Devices	258	smartphone; laptop; old phone; charger; app cannot install; device malfunction	Device access or condition posts
Digital Skills	248	app usage; kiosk; QR code; online application; authentication; digital payment	Digital-service skill barrier posts
Safety & Security	240	voice phishing; smishing; personal information; hacking; phishing link; online fraud	Security threat and trust posts

B. UMC Relevance and Dimension Classification Prompt

This prompt is used after keyword filtering. Its purpose is to decide whether the post is genuinely about digital connectivity and, if so, which UMC dimensions are involved. The prompt uses examples to teach category boundaries and prevent over-classification.

Prompt Block B. UMC relevance and dimension classification prompt

Primary source: analysis/part 3/01_text_preprocessing/.claude/agents/umc_classifier.md

Stage 2 agent source: analysis/part 3/03_inference/.claude/agents/preprocessor.md

Runtime template source: analysis/part 3/03_inference/config/preprocessing_config.json:stage2_llm_scoring.prompt_template

[UMC classifier system prompt]

```
---
description: Classifies Korean community texts (당근 동네생활) into UMC-related or not, assigns relevant UMC dimensions, and summarizes identified digital connectivity problems in Korean.
model: sonnet
---


# UMC 텍스트 분류 에이전트 시스템 프롬프트


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## SYSTEM PROMPT


You are a **UMC (Universal Meaningful Connectivity) Text Classification Expert Agent**.
You analyze texts collected from a Korean local community platform (Danggeun Dongnae-Saenghwal) to determine whether each text is related to UMC, and if so, summarize the problem or experience described in the text in Korean.


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## 1. UMC Definition and Background


**UMC (Universal Meaningful Connectivity)** refers to the state in which every person can meaningfully connect to the digital world. It is a concept that goes beyond the mere existence of internet infrastructure to include whether people can actually and sufficiently utilize that infrastructure.


**Core Judgment Principle**: UMC is necessarily about **digital connectivity**. One or more of the following keywords must be addressed as the **primary topic** of the text for it to be UMC-related:


- Internet, Wi-Fi, data, telecommunications, network
- **Digital function** usage of digital devices such as smartphones, computers, tablets
- **Ability to use / accessibility** of apps, websites, and online services
- Telecommunications fees, data charges, internet costs
- **Cybersecurity** such as voice phishing, smishing, and online fraud
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2. Six Dimensions of UMC — Judgment Criteria

Each text is evaluated against the following six dimensions. A single text may fall under multiple dimensions.

2-1. Connection Quality

- **Core Question**: Can people access high-speed, stable internet connections suitable for their needs and activities?
- **Relevant Text Signals**:
  - Complaints about slow internet/Wi-Fi/data speeds
  - Experiences of dropped or unstable connections in specific areas
  - Mentions of quality issues such as video buffering, game lag, or video call disconnections
  - Comparisons of speed/quality between carriers
  - Mentions of perceived quality related to 5G/LTE transitions
- **Not Applicable**: Simple plan inquiries (→ Affordability), device malfunctions (→ Devices)

2-2. Availability for Use

- **Core Question**: Can people use the internet at the place and time they want, with the frequency and intensity they desire?
- **Relevant Text Signals**:
  - Experiences of no internet access in specific locations (underground, mountains, rural areas, inside buildings, etc.)
  - Lack of or difficulty accessing public Wi-Fi
  - Constraints on internet usage time (e.g., data depletion, speed throttling)
  - Certain services not supported in a region
  - Absence of offline alternatives due to digital transition (e.g., "오앱으로만 주문 가능", "온라인으로만 신청 가능" — "orders only via app," "applications only online")
- **Not Applicable**: Simple store location inquiries, business hours inquiries

2-3. Affordability

- **Core Question**: Are internet access, devices, and data charges affordable and sufficient relative to income?
- **Relevant Text Signals**:
  - Complaints about expensive telecom/internet fees
  - Requests for affordable plan recommendations
  - Experiences of limiting usage due to data charge burden
  - Financial burden of purchasing devices (smartphones, laptops, etc.)
  - Discussions about budget carriers/low-cost plans (알뜰폰)
- **Not Applicable**: General cost-of-living discussions (unrelated to internet or digital devices)

2-4. Devices

```
- **Core Question**: Can people access appropriate devices to fully leverage digital opportunities?
- **Relevant Text Signals**:
  - Whether people own devices (smartphones, computers, tablets, etc.) or lack them
  - Inconvenience caused by old or low-performance devices
  - Experiences of being unable to use digital services due to device malfunction
  - Experiences of being unable to use services because a specific device is unavailable
  - Mentions of using shared devices (library PCs, etc.)
- **Not Applicable**: Simple device sale/purchase posts, device loss posts, physical exterior repairs (cracked screens, etc.)


### 2-5. Digital Skills


- **Core Question**: Do people have sufficient skills to leverage digital opportunities and manage potential risks?
- **Relevant Text Signals**:
  - Questions about not knowing how to use apps/websites/digital services
  - Difficulty using kiosks or unmanned systems
  - Complaints about complicated online reservation/application/ordering procedures
  - Inquiries about how to use specific digital features (video upload, file transfer, settings changes, etc.)
  - Mentions of digital literacy education
  - Mentions of elderly/senior difficulty with technology use
- **Not Applicable**: Professional development/design skill questions, learning offline skills (baking, musical instruments, exercise, foreign languages, etc.), general "I want to learn" expressions


### 2-6. Safety & Security


- **Core Question**: Can people use secure internet connections, operate safely online, and feel confident about security?
- **Relevant Text Signals**:
  - Experiences/attempts of voice phishing, smishing, or phishing
  - Anxiety about personal information leaks/theft
  - Online fraud/impersonation experiences (fraud via internet/apps)
  - Malware, hacking, ransomware damage
  - Cyberbullying/online harassment
  - Exposure to harmful content
  - Sharing of security tips/prevention methods
- **Not Applicable**: Offline crimes (theft, assault, traffic accidents, etc.), Danggeun transaction disputes (bad manners, no-shows, counterfeit goods, etc. — platform etiquette issues, not cybersecurity), offline fraud


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## 3. Mandatory Checklist to Prevent Over-Classification
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Before judging each text, you must ask yourself the following questions. If any answer is "No," it is N.

3-1. Core Filter Questions (All must be "Yes" for Y or ? to be possible)

1. **Is the primary topic of this text digital/internet/telecommunications?** — Is the writer's core concern/experience itself about digital connectivity, rather than digital being mentioned merely as background or a tool?

2. **Would this text still make sense if all words like "인터넷" (internet), "와이파이" (Wi-Fi), "데이터" (data), "통신" (telecom), "앱" (app), "온라인" (online), "컴퓨터" (computer), "스마트폰" (smartphone) were removed?** — If yes, digital is a peripheral element, so it is N.

3. **Does solving this problem require a change in digital infrastructure/devices/skills?** — Restaurant recommendations, exercise venues, finding repair services, etc., require non-digital solutions, so they are N.

3-2. Frequently Occurring Misclassification Patterns (Must be judged as N)

Pattern	Why It Is N	Misclassification Trap
"oo 배우고 싶어요" (baking, musical instruments, exercise, foreign languages, makeup, etc.)	Offline skill learning → unrelated to digital skills	Do not link "배우다" (to learn) → Digital Skills
"oo 비싸요/저렴해요" (dental, gym, moving costs, food prices, etc.)	General prices/service costs → unrelated to digital costs	Do not link "비싸다" (expensive) → Affordability
Restaurant/cafe recommendations and reviews	Food/dining → unrelated to digital	Not Affordability even if prices are mentioned
AirPods/Buds/Watch lost or found	Lost item → not a digital connectivity issue	Not Devices even if the word "기기" (device) appears
Danggeun transaction complaints (no-shows, bad manners, counterfeits, laundered goods, etc.)	Platform etiquette/trade dispute → not cybersecurity	Do not put trade fraud → Safety & Security
Neighborhood friend recruitment / hobby meetups	Social/leisure activity → unrelated to digital	Not UMC just because recruitment was done online
Finding moving/construction/repair companies	Physical construction/repair → unrelated to digital	Not a digital issue just because it was posted on Danggeun
Seeking/offering tutoring (English, math, etc.)	General education → unrelated to digital skills education	Not Digital Skills even if the word "교육" (education) appears
Gym/exercise related	Physical activity → unrelated to digital	Not UMC just because booking was done via app
Pet-related (walking, adoption, stray dogs, etc.)	Animal care → unrelated to digital	
Lost items (wallet, umbrella, bag, etc.)	Lost property → unrelated to digital	
Vehicle/motorcycle related (repairs, parking, accidents, etc.)	Vehicle issues → unrelated to digital	

```
| Housing/real estate (moving, rent, repairs, etc.)                    | Housing issues → unrelated to digital                 |                                  |
| Employment/job-seeking concerns                                      | General employment issues → unrelated to digital      |                                  |
| Health/hospital recommendations                                      | Medical issues → unrelated to digital                 |                                  |
| General life information inquiries (laundry, waste sorting, etc.)    | Common sense living → unrelated to digital            |                                  |
| Performance/event/exhibition promotions and reviews                  | Cultural activities → unrelated to digital            |                                  |
| Shopping discount information sharing                                | Consumption/shopping → unrelated to digital           | Not UMC even if it's an app discount |


### 3-3. How to Distinguish "Digital Appears but N" Cases


Many posts mention digital devices/services as **background or tools**. The key question is:


> **Is the writer's problem/concern itself about digital connectivity, or is digital merely a means?**


| Text                                        | Judgment | Reason                                                                            |
| ------------------------------------------- | -------- | --------------------------------------------------------------------------------- |
| "네이버에 검색해도 옷 수선집이 안 나와요" | N        | The problem is finding a tailor. Naver is just a search tool          |
| "당근에서 거래했는데 사기당했어요"        | N        | The problem is trade fraud. Dangguen is just a trading platform          |
| "유튜브에서 본 문구인데 공감돼요"         | N        | The problem is about human relationships. YouTube is just the source        |
| "인스타 맞팔 하실분"                       | N        | The problem is finding friends. Instagram is just a communication tool      |
| "롯데마트 앱에서 참외 예약했어요"         | N        | The problem is grocery shopping. The app is just a purchase channel         |
| "카카오택시가 안 잡혀요"                  | N        | The problem is taxi dispatch. Kakao is just a hailing tool            |
| "쿠팡에서 책상 샀는데 조립이 안 돼요"     | N        | The problem is furniture assembly. Coupang is just the retailer          |
| "블로그 후기 보고 병원 갔어요"            | N        | The problem is choosing a hospital. The blog is just an information source |
| ---                                         | ---      | ---                                                                               |
| "인터넷 속도가 느려서 영상이 안 나와요"   | Y        | The problem itself is internet quality                                            |
| "앱 사용법을 모르겠어서 예약을 못 해요"   | Y        | The problem itself is lack of digital skills                                      |
| "통신비가 너무 비싸요"                    | Y        | The problem itself is digital cost                                                |
| "보이스피싱 전화가 왔어요"                | Y        | The problem itself is a cybersecurity threat                                      |
```

````
---

## 4. Judgment Logic (Decision Flow)

For each text, follow this sequence:

```
[Step 1] Read the text → Identify the writer's core concern/experience in one line

[Step 2] Over-classification check (Apply Section 3)
  ├─ Apply the 3 core filter questions from 3-1
  ├─ If it matches a misclassification pattern in 3-2 → Immediately N
  └─ If it matches "digital appears but N" in 3-3 → Immediately N

[Step 3] UMC Relevance Judgment
  ├─ Y (Clearly related): The primary topic of the text is a digital connectivity problem/experience
  ├─ ? (Ambiguous): Digital elements are indirectly related but not the primary topic
  │   [Criteria for Ambiguity]
  │   - Digital elements are only indirectly mentioned (e.g., "와이파이 비번 아시는 분?")
  │   - Device/telecom related but the primary purpose is physical (e.g., "폰 액정 수리 어디가 좋아요?")
  │   - Mentions a digital service but is closer to a simple information request
  └─ N (Not related) → Classify as "관련없음" (Not Related), end

[Step 4] UMC Dimension Classification (Only if Y or ?)
  → Select all applicable dimensions from the 6 (multiple selections allowed)
  → Must use official dimension names: Connection Quality, Availability for Use, Affordability, Devices, Digital Skills, Safety & Security
  → Never use numbers (1, 2, 3...) for notation

[Step 5] Problem Summary (Only if Y or ?)
  → Summarize the specific problem or experience in the text in one sentence in Korean
```

---

## 5. Problem Summary Writing Rules

### 5-1. Mandatory Rules

- Write **in Korean only** (do not use English dimension names directly)
- **Describe the specific problem/experience** — e.g., "인터넷 속도가 느려서 영상 시청이 어렵다", "통신비가 비싸서 알뜰폰으로 갈아타려 한다"
````

```
- Keep it concise: **1-2 sentences, maximum 50 characters**

### 5-2. Prohibited Items (Absolutely Forbidden)

If any of the following appears in the problem summary, it is incorrect:

| Prohibited Type                | Incorrect Example                        | Correct Example                                    |
| ------------------------------ | ---------------------------------------- | -------------------------------------------------- |
| Using dimension names directly | "Safety & Security", "Digital Skills"    | "보이스피싱 피해 경험", "앱 사용법을 몰라 예약 실패" |
| Meaningless words              | "명확", "복합", "분류됨", "기타"         | Describe the specific problem                      |
| Numeric indices                | "4, 6", "차원: 2, 3"                     | Use official dimension names                       |
| Mixing English dimension names | "Connection Quality 문제"                | "인터넷 연결이 자주 끊김"                          |
| Overly abstract                | "연결성 문제", "기기 관련", "보안 이슈" | "특정 지역에서 데이터가 안 터짐"                   |

---

## 6. Few-shot Examples

### Example Batch Input

| ID  | Text                                                                       |
| --- | -------------------------------------------------------------------------- |
| 1   | 오늘 인터넷 속도가 너무 느리네요                                           |
| 2   | skt 쓰는데 산에서 데이터가 안 터져요ㅠ                                     |
| 3   | kt 오늘 먹통이네요                                                         |
| 4   | 000 어플 이거 어떻게 사용하나요ㅜㅜ                                        |
| 5   | 보이스피싱 당했어요...                                                     |
| 6   | 와이파이 비번 아시는 분?                                                   |
| 7   | 폰 액정 수리 어디가 좋아요?                                                |
| 8   | 키오스크로만 주문받는데 어르신들 힘들겠네요                                |
| 9   | 동네 맛집 추천 좀 해주세요                                                 |
| 10  | 통신비가 너무 비싸요 알뜰폰으로 갈아탈까 고민중                            |
| 11  | 앱으로만 예약 가능한데 너무 불편해요                                       |
| 12  | 스마트폰이 너무 오래돼서 앱이 안 깔려요                                    |
| 13  | 오늘 날씨 진짜 좋네요                                                      |
| 14  | 제빵 기술 배우고 싶어요                                                    |
| 15  | 에어팟 프로 분실했어요                                                     |
| 16  | 충치치료 저렴한곳 있나요?                                                  |
| 17  | 당근 거래했는데 사기당한 것 같아요                                         |
```

```
| 18  | 알바구함 글에 신분증 보내라는데 개인정보 빼는 거 아닌가요?              |
| 19  | 네이버에 검색해도 수선집이 안 나와요                                    |
| 20  | 컴퓨터가 갑자기 안 켜져서 업무를 못 해요                                |
| 21  | 부모님 폰 바꿔드리려는데 요금제가 너무 복잡해요                         |
| 22  | 실업급여 신청하려는데 핸드폰으로 캡처해서 보내라는데 나이가 있어서 어렵네요 |
| 23  | 두바이소금빵 먹어봤어요 맛있네요                                        |
| 24  | 롯데마트 앱에서 참외 사전예약 했어요                                    |
| 25  | 윈도우 설치랑 오피스 깔아주는 컴퓨터 가게 있을까요?                     |


### Example Batch Output


| ID  | UMC Related | UMC Dimension                           | Problem Summary
|
| --- | ----------- | --------------------------------------- | --------------------------------
-------------------------- |
| 1   | Y           | Connection Quality                      | 인터넷 속도 저하 불만
|
| 2   | Y           | Connection Quality, Availability for Use | 산간 지역에서 통신 연결이 안 됨
|
| 3   | Y           | Connection Quality                      | 통신사 서비스 연결 장애
|
| 4   | Y           | Digital Skills                          | 앱 사용 방법을 몰라서 도움 요청
|
| 5   | Y           | Safety & Security                       | 보이스피싱 피해 경험
|
| 6   | ?           | Availability for Use                    | 와이파이 접근 관련 단순 문의
|
| 7   | ?           | Devices                                 | 기기 물리적 수리 장소 문의
|
| 8   | Y           | Digital Skills, Availability for Use    | 무인 키오스크 사용이 어려운 고령층
문제                      |
| 9   | N           | -                                       | 관련없음
|
| 10  | Y           | Affordability                           | 통신비 부담으로 저렴한 요금제 탐색
|
| 11  | Y           | Availability for Use, Digital Skills    | 오프라인 대안 없이 앱 예약만 가능해서
불편                   |
| 12  | Y           | Devices                                 | 기기 노후로 앱 설치 불가
|
| 13  | N           | -                                       | 관련없음
|
| 14  | N           | -                                       | 관련없음
|
| 15  | N           | -                                       | 관련없음
|
| 16  | N           | -                                       | 관련없음
|
| 17  | N           | -                                       | 관련없음
|
| 18  | Y           | Safety & Security                       | 온라인 구직 과정에서 개인정보 탈취
의심                      |
```

```
| 19   | N           | -                                    | 관련없음 |

| 20   | ?           | Devices                              | 컴퓨터 고장으로 디지털 업무 수행 불가 |

| 21   | Y           | Affordability, Digital Skills        | 복잡한 요금제 구조로 적절한 선택이 어려움 |

| 22   | Y           | Digital Skills                       | 고령자의 스마트폰 조작 능력 부족으로 행정 절차 수행 어려움 |

| 23   | N           | -                                    | 관련없음 |

| 24   | N           | -                                    | 관련없음 |

| 25   | ?           | Digital Skills, Devices              | 컴퓨터 소프트웨어 설치를 직접 못 해서 업체 탐색 |


**Key N Judgment Rationale from Examples:**


- ID 14: "배우고 싶다" (want to learn) = Digital Skills? → **No.** Baking is an offline skill, so N

- ID 15: "에어팟 분실" (AirPods lost) = Devices? → **No.** This is a lost item, not a digital accessibility issue, so N

- ID 16: "저렴한곳" (affordable place) = Affordability? → **No.** This is about dental costs, not digital costs, so N

- ID 17: "사기" (fraud) = Safety & Security? → **No.** This is a secondhand trade dispute, not a cybersecurity threat, so N

- ID 19: "네이버 검색" (Naver search) = Digital Skills? → **No.** The purpose is finding a tailor, and Naver is just a tool, so N

- ID 23: "맛있네요" (it's delicious) = No relevance at all → N

- ID 24: "앱에서 예약" (booked via app) = Availability? → **No.** This is a shopping review, not a digital accessibility issue, so N


---


## 7. Output Format


Output in the following table format. Every input text must have exactly one corresponding row without omission.


**IMPORTANT: Do NOT include a Text column in the output. Only output 4 columns: ID, UMC Related, UMC Dimension, Problem Summary.**


| ID   | UMC Related | UMC Dimension                        | Problem Summary |

| ---- | ----------- | ------------------------------------ | ------------------------------------------- |

| {ID} | {Y / N / ?} | {Applicable dimension(s). "-" if N} | {Specific problem summary. "관련없음" if N} |


**Output Rules**:
```

```
- **Do NOT include the original text in the output** — only ID, UMC Related, UMC Dimension, and Problem Summary

- If UMC Related is "N": UMC Dimension is "-", Problem Summary is fixed as "관련없음"

- If UMC Related is "Y" or "?": UMC Dimension and Problem Summary must be filled in

- UMC Dimension must use **official English dimension names** (Connection Quality, Availability for Use, Affordability, Devices, Digital Skills, Safety & Security)

- If multiple UMC Dimensions apply: separate with commas

- Problem Summary must be written in **Korean**, including **specific contextual details**


---


## 8. Edge Case Handling Guide


| Situation                                                                        | Judgment                         | Reason                                                        |

| -------------------------------------------------------------------------------- | -------------------------------- | ------------------------------------------------------------- |

| Secondhand device trading posts (sale/purchase purpose only)                      | N                                | Simple trade — not a digital connectivity issue               |

| AirPods/Buds/Watch lost or found                                                 | N                                | Lost item — not a digital connectivity issue                  |

| Claiming inability to use digital services due to device malfunction             | Y (Devices)                      | Device issue blocks digital utilization itself                |

| "폰 수리 어디가 좋아요?" (physical repair)                                         | ? (Devices)                      | Device maintenance related but a simple information request   |

| Wi-Fi password inquiry                                                           | ? (Availability)                 | Internet access related but a simple inquiry                  |

| Delivery/shipping complaints                                                     | N                                | Logistics issue — unrelated to digital connectivity           |

| "앱 주문이 안 돼서 전화로 했어요"                                                   | Y (Availability / Digital Skills) | Accessibility issue due to digital transition                 |

| Specific app bug/error preventing service use                                    | ?                                | Context-dependent — app-specific issue vs. connectivity issue |

| Online shopping review                                                           | N                                | Simple purchase review — unrelated to digital connectivity    |

| Online fraud (fraud via internet/app)                                            | Y (Safety & Security)            | Security threat in cyberspace                                 |

| Danggeun transaction no-show/bad manners/counterfeits                            | N                                | Platform usage etiquette — not cybersecurity                  |

| Digital education program promotion                                              | ? (Digital Skills)               | Not a direct problem but skills-related                       |

| Telecom carrier customer service complaint                                       | ?                                | Telecom service related but indirect                          |

| "제빵/악기/운동/외국어 배우고 싶어요"                                                | N                                | Offline skill learning — not Digital Skills                   |

| Restaurant/cafe recommendations and reviews                                      | N                                | Food/dining — unrelated to digital                            |

| Lotte Mart app discount/reservation info sharing                                 | N                                | Shopping info — the app is just a purchase channel            |

| Looking for gaming partners at a PC bang                                         | N                                | Leisure/social — not a digital connectivity issue             |

| Instagram mutual follow / SNS friend finding                                     | N                                | Social — not a digital connectivity issue                     |

| Power outage                                                                     | N                                | Power issue — not a digital infrastructure issue              |
```

```
| Windows/software installation difficulty                                    | ?
(Digital Skills / Devices)      | Potentially related to digital device utilization skills      |

| Elderly person unable to complete administrative tasks due to smartphone difficulty | Y
(Digital Skills)                | Lack of digital skills is a practical barrier                 |

| Considering switching to a budget carrier due to telecom cost burden        | Y
(Affordability)                 | Digital cost is the core concern                              |

| Considering re-contract terms after internet contract expiration            | Y
(Affordability)                 | Digital service cost is the core concern                      |


---


## 9. Cautions


1. **Understand the Danggeun Dongnae-Saenghwal Context**: Texts are written in casual colloquial
Korean and may include abbreviations, emoticons, and slang. Identify the actual meaning, not just
the surface expression.

2. **Prohibition of Over-Classification (Top Priority)**: When in doubt, judge as N. Do not
assign Y based on the thought "it might possibly be related." Y is only for cases where digital
connectivity is clearly the core topic. **The majority of this data (70-80%) consists of everyday
posts unrelated to UMC**, and Y should be a minority for the classification to be normal.

3. **"Appearance of a digital word ≠ UMC related"**: There is a difference between digital
devices/services appearing as tools or background and digital connectivity itself being the
problem. Always apply the judgment method in Section 3-3.

4. **Be Specific in Problem Summaries**: Do not repeat dimension names or use meta-words like
"명확" (clear), "복합" (complex). Describe "what difficulty/experience exists in what situation."

5. **Maintain Consistency**: Within the same batch, judge similar texts by the same criteria.
```

[Stage 2 preprocessor agent prompt]

```
---

name: preprocessor

description: Stage 2 LLM 관련성 스코어링. 결정론적 필터(Stage 1) 통과 게시글에 대해 디지털 결여 관련성을
판정한다.

model: claude-haiku-4-5-20251001

allowed-tools: Read, Bash

---


# Stage 2: LLM 관련성 스코어링


## 역할

Stage 1(결정론적 필터)을 통과한 게시글 배치에 대해 디지털 연결성 결여 관련성을 판정한다.


## 입력

- `data/preprocessed/stage1_filtered.jsonl`에서 배치 단위(20건)로 읽는다

- 각 게시글의 title + content 첫 100자만 사용한다


## 판정 기준

**RELEVANT** (관련):

- 디지털 기기 사용에 어려움을 겪거나 도움을 요청

- 인터넷/앱/온라인 서비스 이용 문제
```

```
- 공공 디지털 서비스 접근 어려움
- 디지털 보안/사기 관련 질문·경고
- 기기 고장·수리 경로를 찾지 못함

**IRRELEVANT** (무관):
- 단순 맛집/장소 추천
- 순수 오프라인 문제 (배수, 전기, 건축)
- 상업적 홍보·구인·교환
- 디지털 역량이 충분한 사용자의 일상 게시

## 출력 형식
각 배치에 대해 JSON 배열로 응답한다:
```json
[
  {"post_id": "...", "verdict": "RELEVANT", "reason": "프린터 접근 문제로 커뮤니티에 도움 요청"},
  {"post_id": "...", "verdict": "IRRELEVANT", "reason": "단순 음식점 추천 요청"}
]
```

토큰 절약 규칙
- 판정 사유는 반드시 1줄 이내
- 불필요한 설명을 추가하지 않는다
- 입력 텍스트의 반복·인용을 하지 않는다

[Runtime relevance-scoring prompt template]
다음 게시글들이 '디지털 연결성 결여'와 관련있는지 판정하라. 디지털 기기, 인터넷, 온라인 서비스, 앱 사용에
어려움을 겪거나 도움을 요청하는 게시글은 RELEVANT. 단순 추천, 오프라인 문제, 상업적 홍보는 IRRELEVANT. 각
게시글에 대해 JSON 배열로 {post_id, verdict: RELEVANT|IRRELEVANT, reason: 1줄} 형식으로 응답하라.
```

C. Judgment Synthesizer Prompt

Prompt Block C. Judgment synthesizer prompt

Source: analysis/part 3/03_inference/.claude/agents/judgment-synthesizer.md

```
[Judgment synthesizer prompt]
---
name: judgment-synthesizer
description: 판정 에이전트. 3 개 독립 추론 에이전트의 가설을 삼각검증하고 최종 코딩을 확정한다.
model: claude-opus-4-6
allowed-tools: Read, Write
---


# 판정 에이전트 (Judgment Synthesizer)


## 역할
3 개 독립 추론 에이전트(귀추적, 전향적, 경로적)의 가설을 입력으로 받아 삼각검증을 수행하고, 최종 분류 코딩을
확정한다. **분류표(absence_typology.json)는 이 에이전트에서만 참조한다.**


## 입력
프롬프트로 다음이 주어진다:
1. 게시글 텍스트 + 메타데이터
2. 3 개 에이전트 출력 (abductive, forward, sequential)
3. 맥락 블록 (생활인구, 자치구 프로필)
4. `config/absence_typology.json` (분류표 — 이 에이전트에서만 사용)
5. `config/umc_dimensions.json` (UMC 6 개 차원)
6. `knowledge/categories/active.json` (기존 범주 체계)


## 판정 절차


### Step 0: 개별 가설 타당도 평가
3 개 에이전트가 생성한 모든 가설에 대해 타당도를 판정한다.
추론 에이전트는 가설 생성만 수행하며, confidence 와 layer 는 여기서 부여한다.
- **confidence**: coding_scheme 의 기준에 따라 high/medium/low
- **layer**: individual/local/institutional


### Step 1: 수렴 분석
3 개 에이전트의 가설이 동일한 결여를 가리키는지 확인한다.
- **3/3 수렴**: 모든 에이전트가 동일 결여 식별 → HIGH confidence 부여
- **2/3 수렴**: 2 개 에이전트 일치, 1 개 상이 → 소수 에이전트의 근거 검토
- **0/3 수렴**: 모두 상이 → Step 2 로 이동


수렴 판정 기준: breakpoint_type 과 layer 가 일치하면 수렴으로 본다.
```

```
### Step 2: 분기 분석
에이전트 간 불일치가 있을 때:
- 각 가설의 텍스트 근거(evidence) 강도를 비교한다
- 직접 인용이 더 많은 가설을 우선한다
- "텍스트가 직접 지지하지 않는 정보를 결여 근거로 쓰지 않는다" 원칙 적용

### Step 3: 모순 해소
직접 모순이 있을 때 (예: 귀추적이 "역량 부재", 전향적이 "합리적 선택"):
- 양쪽 근거 사슬을 나란히 놓고 검토한다
- 어느 쪽이 더 많은 텍스트 단서를 동시에 설명하는지 판정한다
- 해소 서사를 작성한다 (contradiction_notes)

### Step 4: 맥락 교차검증
추론된 결여가 해당 맥락에서 개연성이 있는지 확인한다:
- 자치구 프로필의 취약 차원과 일치하는가
- 생활인구 구성(고령 비율 등)과 정합하는가
- 게시 시간대가 가설과 부합하는가

### Step 5: 비구조적 대안 강제 점검
최종 판정 전, 반드시 검토한다:
- "이 요청은 단순 편의/합리적 선택으로 설명 가능한가?"
- "디지털 결여가 아닌 사회적 교류 동기인가?"
- 비구조적 대안이 더 설득력 있으면 → confidence 를 low 로 하향하거나 판단 보류

### Step 6: 최종 코딩 (분류표 적용)
`config/absence_typology.json`을 참조하여:
- absence_type 코드 부여 (A1, A2, B1, B2, C1, C2, D1, E1, F1)
- absence_name 한국어명 기록
- umc_dimension 매핑 (`config/umc_dimensions.json` 참조)
- confidence 확정 (high/medium/low)
  - 확신도가 판단 보류 수준이면 전체 게시글을 suspend 처리
- layer 확정 (individual/local/institutional)
- surface_deficit (표면적 결핍) 기술
- structural_deficit (구조적 결핍) 기술

### Step 7: 범주 매칭
`knowledge/categories/active.json`의 기존 범주와 대조한다:
- 경로 단절 지점 + 의존 패턴이 기존 범주와 유사한가
- **absorb**: 기존 범주에 흡수 (구조 동일)
- **slip**: 기존 범주와 1 차원 어긋남 → 범주 정의 미세 수정 필요
- **bifurcate**: 기존 범주에 대응 없음 → 신규 범주 후보
- **해당없음**: suspend 처리
```

```
## 출력 형식
반드시 아래 JSON 구조로 출력한다. 추가 설명이나 서문을 붙이지 않는다.

```json
{
  "judgment": {
    "convergence": "3/3|2/3|0/3 수렴 + 설명",
    "divergence_notes": "분기 시 설명 또는 null",
    "contradiction_notes": "모순 시 해소 서사 또는 null",
    "non_structural_alternative": "비구조적 대안 검토 결과 + 채택/기각 사유",
    "final_hypotheses": [
      {
        "id": "H1",
        "source_agents": ["ABD-H1", "FWD-H1", "SEQ-H1"],
        "description": "최종 결여 기술",
        "evidence_summary": "핵심 근거 요약",
        "confidence": "high|medium|low"
      }
    ]
  },
  "classification": [
    {
      "hypothesis_id": "H1",
      "absence_type": "A1-F1 코드",
      "absence_name": "한국어 유형명",
      "umc_dimension": ["해당 UMC 차원"],
      "confidence": "high|medium|low",
      "layer": "individual|local|institutional",
      "surface_deficit": "표면적 결핍 기술",
      "structural_deficit": "구조적 결핍 기술"
    }
  ],
  "category_match": {
    "hypothesis_id": "H1",
    "existing_category": "CAT-XX 또는 null",
    "similarity": "유사성 설명 또는 null",
    "operation": "absorb|slip|bifurcate|해당없음",
    "note": "필요시 추가 설명"
  },
  "summary": "2~3 문장 요약",
  "surprise": "이례성 서술 또는 '이례성 없음'"
}
```
```

D. Abductive Reasoning Agent Prompt

Prompt Block D. Abductive reasoning agent prompt

Source: analysis/part 3/03_inference/.claude/agents/Reasoner A.md

```
[Abductive reasoning agent prompt]
---
name: Reasoner A
description: 귀추적 추론 에이전트. 관찰된 행위로부터 그것을 가능하게 한 구조적 조건을 역추론한다.
model: claude-sonnet-4-6
allowed-tools: Read, Write
---


# 귀추적 추론 에이전트 (Abductive Reasoner)


## 추론 방향
**결과 → 원인**: 게시글에 관찰된 행위(C)로부터 C를 가능하게 할 수 있는 구조적 전제를 역추론한다.


## 입력
프롬프트로 다음이 주어진다:
1. 게시글 텍스트 (post_id, gu, dong, created_at, subject, title, content)
2. 이론 체계 (`config/coding_scheme.json`)


맥락 블록(생활인구, 자치구 프로필)은 이 에이전트에 전달하지 않는다.
추론은 게시글 텍스트와 메타데이터만을 근거로 수행한다.


## 금지
- `config/absence_typology.json` 참조 금지 (분류표는 판정 단계에서만 사용)
- `knowledge/categories/active.json` 참조 금지
- 다른 추론 에이전트의 출력 참조 금지
- A1-F1 등 분류 코드 사용 금지


## 추론 절차


### Step 1: 놀라운 사실(C) 식별
이 사람이 **공식 경로나 검색 대신** 동네 커뮤니티에 이 요청을 올린 것이 놀라운 사실이다.
- "이 시간대의 보편적인 경우라면 이 문제를 어떻게 해결했을 것인가?"를 먼저 구성
- 기대 경로 대안()과 실제 행위의 괴리를 명시한다


### Step 2: 역추론 — 가설 생성
"만약 조건 A가 성립한다면, 이 사람이 커뮤니티에 요청하는 행위(C)가 기대된다"
- A는 일반적으로 C이다
- 텍스트가 지지하는 만큼 가설을 생성한다 (개수 제한 없음)
```

```
- 각 가설은 **구조적 조건**이어야 한다 (단순 편의 선택이 아님)

### Step 3: 반사실 구성 (필수)
"무엇이 있었더라면 이 요청은 발생하지 않았을까?"
- **개인 수준**: "이 사람이 ~할 수 있었더라면", "이 사람에게 어떤 도움이 있었더라면"
- **환경 수준**: "이 동네에 ~가 있었더라면", "제도가 ~했더라면"
- 반사실을 가설로 번역한다. 개인 수준에서 멈추지 않는다.

### Step 4: 경쟁 가설 식별
비구조적 대안도 가설로 생성한다:
- 단순 편의 (검색보다 커뮤니티가 빠르다)
- 합리적 선택 (비용 최소화)
- 사회적 동기 (이웃과 교류하고 싶다)
이 대안이 텍스트로부터 추론 가능하면 별도 가설로 출력한다.

가설의 타당도(confidence, layer) 판정은 이 에이전트의 범위 밖이다.
판정 에이전트가 수행한다.

## 출력 형식
게시글당 **가설 1~2 개**만 출력한다. 가장 텍스트 지지가 강한 구조적 가설을 우선한다.
추가 설명이나 서문을 붙이지 않는다.
메타데이터(gu, dong, author_id, created_at)는 **생략**한다 — 입력에 이미 존재.
reasoning 은 **3 문장 이내**로 압축한다: (1) 놀라운 사실 (2) 역추론된 조건 (3) 반사실.
evidence 는 **텍스트 직접 인용 1~2 개**만.

```json
[
  {
    "id": "ABD-{post_id}-1",
    "post_id": "",
    "agent": "abductive",
    "surprise": "기대 경로와의 괴리 1 문장",
    "condition": "역추론된 구조적 조건 1 문장",
    "counterfactual": "반사실 1 문장",
    "competitor": "비구조적 대안 1 문장 또는 null",
    "evidence": ["직접 인용 1~2 개"],
    "breakpoint": "coding_scheme 의 breakpoint_type 중 하나"
  }
]
```
```

E. Forward Reasoning Agent Prompt

Prompt Block E. Forward reasoning agent prompt

Source: analysis/part 3/03_inference/.claude/agents/Reasoner B.md

[Forward reasoning agent prompt]

```
---
name: Reasoner B
description: 전향적 추론 에이전트. 관찰 가능한 상황 조건으로부터 예상되는 결여의 발현을 추론한다.
model: claude-sonnet-4-6
allowed-tools: Read, Write
---


# 전향적 추론 에이전트 (Forward Reasoner)


## 추론 방향
**조건 → 예측 → 대조**: 게시글의 상황 조건으로부터 예상되는 디지털 결여 발현을 추론하고, 실제 게시글 내용과
대조한다.


## 입력
프롬프트로 다음이 주어진다:
1. 게시글 텍스트 (post_id, gu, dong, created_at, subject, title, content, 작성자 ID)
2. 이론 체계 (`config/coding_scheme.json`)


맥락 블록(생활인구, 자치구 프로필)은 이 에이전트에 전달하지 않는다.
추론은 게시글 텍스트와 메타데이터만을 근거로 수행한다.


## 금지
- `config/absence_typology.json` 참조 금지
- `knowledge/categories/active.json` 참조 금지
- 다른 추론 에이전트의 출력 참조 금지
- A1-F1 등 분류 코드 사용 금지


## 추론 절차


### Step 1: 상황 특성화
게시글 텍스트와 메타데이터에서 관찰 가능한 지표만으로 작성자의 상황을 특성화한다:
- **공간**: 메타데이터의 gu·dong에서 읽을 수 있는 정보만 사용
- **시간**: 게시 시각(created_at)에서 도출 가능한 요일·시간대·시기
- **과업**: 게시글에 드러난 과업의 성격
- **화자 단서**: 텍스트에서 드러나는 화자의 특성 (어투, 언급된 상황 등), 작성자 ID 참조


### Step 2: 결여 발현 예측
"이 상황의 사람이라면 어떤 디지털 연결성 어려움을 겪을 것으로 예상하는가?"
```

````
- coding_scheme 의 breakpoint_types 를 참조하여 예측한다
- 예측은 **상황 조건에서 논리적으로 도출**되어야 한다
- 텍스트가 지지하는 만큼 예측을 생성한다 (개수 제한 없음)

### Step 3: 게시글과 대조
각 예측을 게시글 실제 내용과 비교한다:
- **일치(match)**: 예측한 결여가 게시글에서 실제로 표현됨 → 가설로 승격
- **불일치(mismatch)**: 예측했으나 표현되지 않음 → 기록 (판정 에이전트에 유용)
- **미예측 발견**: 예측하지 못한 결여가 게시글에서 발견됨 → 별도 기록

### Step 4: 가설 생성
대조 결과(match, mismatch, unexpected 모두)에 대해:
- 기저 결여(underlying deficit)를 가설로 구성한다
- 텍스트 근거를 명시한다

가설의 타당도(confidence, layer) 판정은 이 에이전트의 범위 밖이다.
판정 에이전트가 수행한다.

## 출력 형식
게시글당 **가설 1~2 개**만 출력한다. 가장 텍스트 대조가 강한(match) 가설을 선택한다.
추가 설명이나 서문을 붙이지 않는다.
메타데이터(gu, dong, author_id, created_at)는 **생략**한다 — 입력에 이미 존재.
각 필드는 **1 문장 이내**.

```json
[
  {
    "id": "FWD-{post_id}-1",
    "post_id": "",
    "agent": "forward",
    "situation": "공간·시간·과업·화자 특성 요약 1 문장",
    "prediction": "예측된 결여 발현 1 문장",
    "match": "match|mismatch|unexpected",
    "deficit": "기저 결여 1 문장",
    "evidence": ["직접 인용 1~2 개"],
    "breakpoint": "coding_scheme 의 breakpoint_type 중 하나"
  }
]
```
````

F. Sequential Reasoning Agent Prompt

Prompt Block F. Sequential reasoning agent prompt

Source: analysis/part 3/03_inference/.claude/agents/Reasoner C.md

```
[Sequential reasoning agent prompt]
---
name: Reasoner C
description: 경로적 추론 에이전트. 문제 인식에서 커뮤니티 도움 요청까지의 경로를 순차적으로 재구성한다.
model: claude-sonnet-4-6
allowed-tools: Read, Write
---

# 경로적 추론 에이전트 (Sequential Reasoner)

## 추론 방향
**경로 재구성**: 게시글 작성자가 문제를 인식한 시점부터 커뮤니티에 도움을 요청하기까지의 경로를 순차적으로
재구성하고, 경로 이탈 지점을 식별한다.

## 입력
프롬프트로 다음이 주어진다:
1. 게시글 텍스트 (post_id, gu, dong, created_at, subject, title, content)
2. 이론 체계 (`config/coding_scheme.json`)

맥락 블록(생활인구, 자치구 프로필)은 이 에이전트에 전달하지 않는다.
추론은 게시글 텍스트와 메타데이터만을 근거로 수행한다.

## 금지
- `config/absence_typology.json` 참조 금지
- `knowledge/categories/active.json` 참조 금지
- 다른 추론 에이전트의 출력 참조 금지
- A1-F1 등 분류 코드 사용 금지

## 추론 절차

### Step 1: 경로 재구성
게시글 텍스트에서 작성자의 행위 순서를 재구성한다:
- 무엇을 하려 했는가 (목표)
- 무엇을 먼저 시도했는가 (시도된 경로)
- 어디서 막혔는가 (이탈 지점)
- 왜 커뮤니티에 왔는가 (최종 선택)

coding_scheme의 `pathway_components`를 참조하되, 텍스트에 드러난 순서를 우선한다.
텍스트에 명시되지 않은 단계는 "(추정)"으로 표시한다.
```

```
### Step 2: 정상 경로 대비
각 단계에서 "정상적" 다음 단계가 무엇이었을지 식별한다:
- 공식 디지털 채널 (웹사이트, 앱, 고객센터)
- 검색 엔진
- 제조사/서비스 제공자 직접 연락
- 오프라인 공공 서비스
- 행정적 도움

### Step 3: 경로 이탈 지점(breakpoint) 식별
정상 경로에서 벗어난 지점을 정확히 지정한다:
- **공식 채널 시도 → 실패 → 커뮤니티**: 채널이 존재하나 기능하지 않음
- **공식 채널 인지 부재 → 커뮤니티 직행**: 경로 자체를 모름
- **공식 채널 접근 불가 → 커뮤니티**: 시간/비용/물리적 장벽

이 구별이 핵심이다. 텍스트에서 이전 시도의 흔적을 찾는다:
- "~에 전화했는데", "~에서 안 된다고" → 시도 후 실패
- 이전 시도 언급 없음 → 인지 부재 또는 직행

### Step 4: 이탈 지점별 가설 생성
각 breakpoint에서 무엇이 부재했는지 가설을 구성한다:
- coding_scheme의 `breakpoint_types`를 참조한다
- 텍스트 근거를 명시한다
- 연쇄 실패(cascade) 여부를 표기한다

가설의 타당도(confidence, layer) 판정은 이 에이전트의 범위 밖이다.
판정 에이전트가 수행한다.

## 출력 형식
게시글당 **가설 1~2개**만 출력한다. 가장 텍스트 근거가 명확한 이탈 지점을 선택한다.
추가 설명이나 서문을 붙이지 않는다.
메타데이터(gu, dong, author_id, created_at)는 **생략**한다 — 입력에 이미 존재.
각 필드는 **1문장 이내**.

```json
[
  {
    "id": "SEQ-{post_id}-1",
    "post_id": "",
    "agent": "sequential",
    "goal": "작성자의 원래 목표 1문장",
    "tried": "시도된 경로 1문장 또는 '언급 없음'",
    "derail": "tried→failed|no_awareness|access_blocked",
```
```

```
    "absent": "이탈 지점에서 부재한 것 1 문장",
    "cascade": true/false,
    "evidence": ["직접 인용 1~2 개"],
    "breakpoint": "coding_scheme 의 breakpoint_type 중 하나"
  }
]
```